



FibreFlow

COMPLETE USER MANUAL

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Application: FibreFlow (fibreflow.app)

Scope: All Modules

Classification: Internal Use

FibreFlow — Complete User Manual

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How to Use This Manual

This manual covers every module in FibreFlow, organized by the sidebar navigation sections. Each chapter includes:

- **Overview** — What the module does and who uses it
- **Navigation** — How to reach the module from the sidebar
- **Features** — Detailed walkthrough of each feature with screenshots
- **Workflows** — Step-by-step procedures for common tasks

- **Troubleshooting** — Solutions for common issues

Conventions used in this manual:

Convention	Meaning
Bold text	UI element names (buttons, tabs, labels)
<code>Code text</code>	Technical values, field names, or system identifiers
> Tip:	Helpful shortcut or best practice
> Important:	Critical information — read carefully
<i>Italic text</i>	Figure captions below screenshots

Role Badges: Where features are restricted by role, you will see tags like **(Admin only)** or **(Manager+)**.

1. Getting Started

1.1 Accessing FibreFlow

FibreFlow is a web application accessible from any modern browser (Chrome, Firefox, Edge, Safari).

Environment	URL	Purpose
Production	app.fibreflow.app	Live system for daily use
Staging	vf.fibreflow.app	Pre-release testing
Development	dev.fibreflow.app	Feature development

1. Open your browser and navigate to **app.fibreflow.app**
2. Enter your email address and password
3. Click **Sign In**
4. You will be redirected to the Dashboard

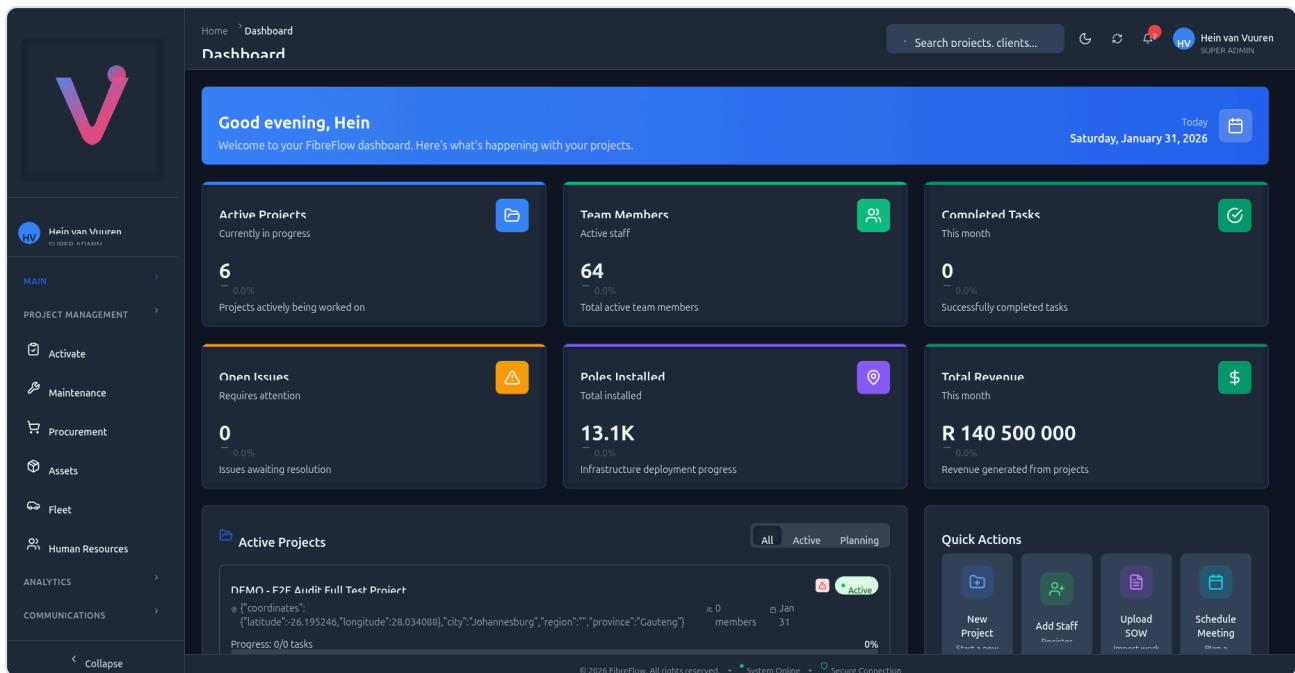


Figure 1.1: FibreFlow login page

Tip: If you forget your password, contact your system administrator to reset it.

1.2 The FibreFlow Interface

After logging in, you will see the main interface consisting of:

- **Sidebar** (left) — Navigation menu organized into sections
- **Header** (top) — Search bar, notifications, user menu, dark mode toggle
- **Content Area** (center) — The active page content
- **Tabs** (within modules) — Horizontal tab navigation within each module

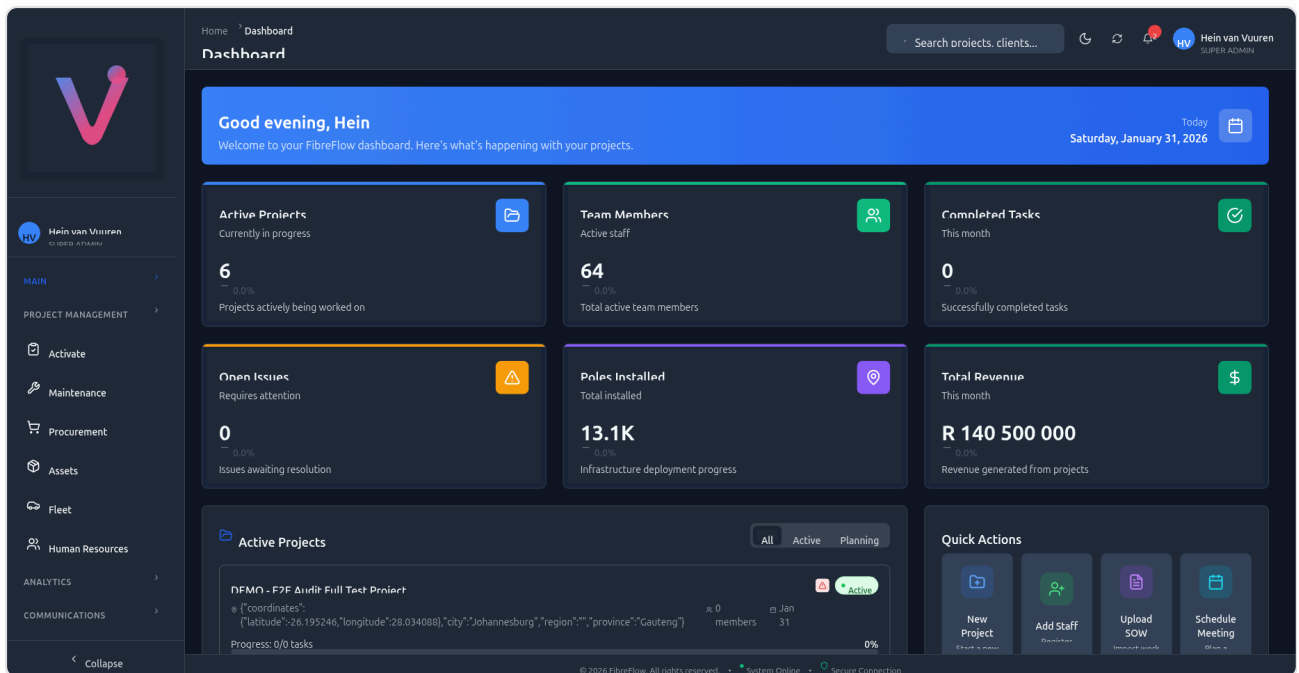


Figure 1.2: FibreFlow main interface showing sidebar, header, and content area

1.3 Navigation Structure

The sidebar is organized into **11 sections**, each representing a major area of the application:

Section	Icon	Purpose
Main	Home	Dashboard, Meetings, Action Items
Project Management	Folder	Projects, Clients, Contractors
Activate	CheckCircle	QA photo review and DR processing
Maintenance	Wrench	Maintenance ticket management
Procurement	ShoppingCart	BOQ, RFQ, PO, inventory
Assets	Package	Equipment and asset tracking
Fleet	Truck	Vehicle and driver management
Human Resources	Users	Staff directory and compliance
Analytics	BarChart	KPIs, reports, dashboards
Communications	MessageSquare	WhatsApp, meetings, notifications
System	Settings	Admin tools, data sync, settings

Click a section header to expand or collapse it. Sections with a direct link (like Activate or Maintenance) navigate directly to that module.

1.4 User Roles & Permissions

FibreFlow uses Role-Based Access Control (RBAC). Your role determines which modules, pages, and actions you can access.

Role	Description	Typical Access
Super Admin	Full system control	Everything — all modules, settings, system admin
Admin	Organization admin	All modules except system internals
Manager	Department/project lead	Projects, procurement, reports, team management
Technician	Field worker	Assigned tasks, check-ins, limited views
Viewer	Read-only observer	View dashboards and reports, no edit access
Contractor	External partner	Contractor portal, assigned project views

Important: *If you cannot see a menu item or feature described in this manual, your role may not have access. Contact your administrator to request additional permissions.*

1.5 Common UI Patterns

FibreFlow uses consistent patterns across all modules:

Search Bars — Type to filter lists in real time. Most search bars filter across multiple fields (name, ID, description).

Filters — Dropdown or toggle filters to narrow results by status, date, category, or other criteria.

Tab Navigation — Horizontal tabs within modules let you switch between sub-sections without leaving the page.

Modal Dialogs — Forms and details often open in overlay modals. Click outside or press Escape to close.

Toast Notifications — Success/error messages appear briefly in the corner of the screen.

Dark Mode — Toggle dark mode from the header menu (user icon → Theme toggle). All modules support dark mode.

2. Main — Dashboard, Meetings & Action Items

2.1 Dashboard

The Dashboard is your landing page after login. It provides a high-level overview of your organization's operations.

Navigation: Sidebar → **Main** → **Dashboard** (or click the FibreFlow logo)

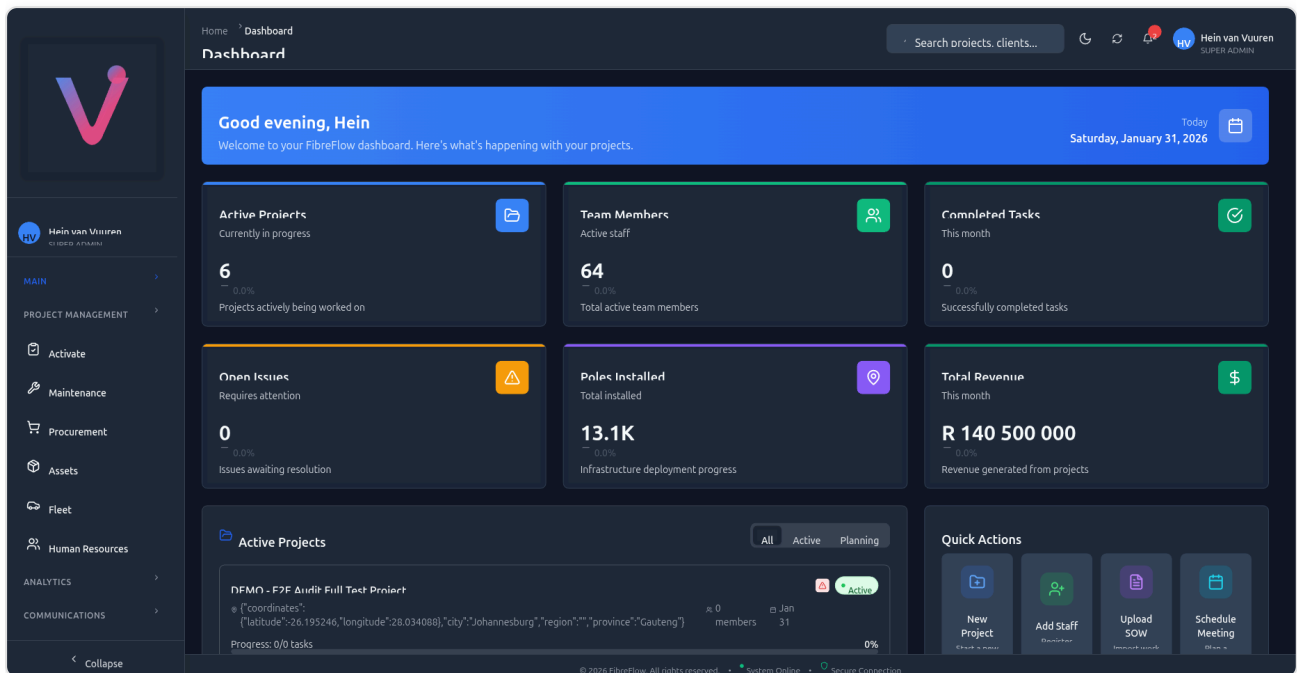


Figure 2.1: Main Dashboard showing project overview cards, activity feed, and quick actions

STATISTICS CARDS

The top row displays key metrics:

Card	Description
Active Projects	Number of currently active projects
Total Staff	Number of staff members in the system
Open Tickets	Maintenance tickets requiring attention
Pending QA	Activate reviews waiting for QA processing

PROJECT OVERVIEW

A grid of project cards showing each project's:

- Name and client
- Status (Active, Planning, On Hold, Completed)
- Progress percentage
- Key dates

RECENT ACTIVITY FEED

A chronological stream of recent actions across the system — ticket updates, project changes, new submissions, and more.

QUICK ACTIONS

Shortcut buttons for common tasks:

- **Create Project** — Start a new project
- **Create Ticket** — Open a maintenance ticket
- **Import SOW** — Upload a Scope of Work document
- **View Reports** — Navigate to analytics

2.2 Meetings

The Meetings module integrates with Fireflies.ai to automatically sync meeting transcripts, summaries, and action items.

Navigation: Sidebar → Main → Meetings

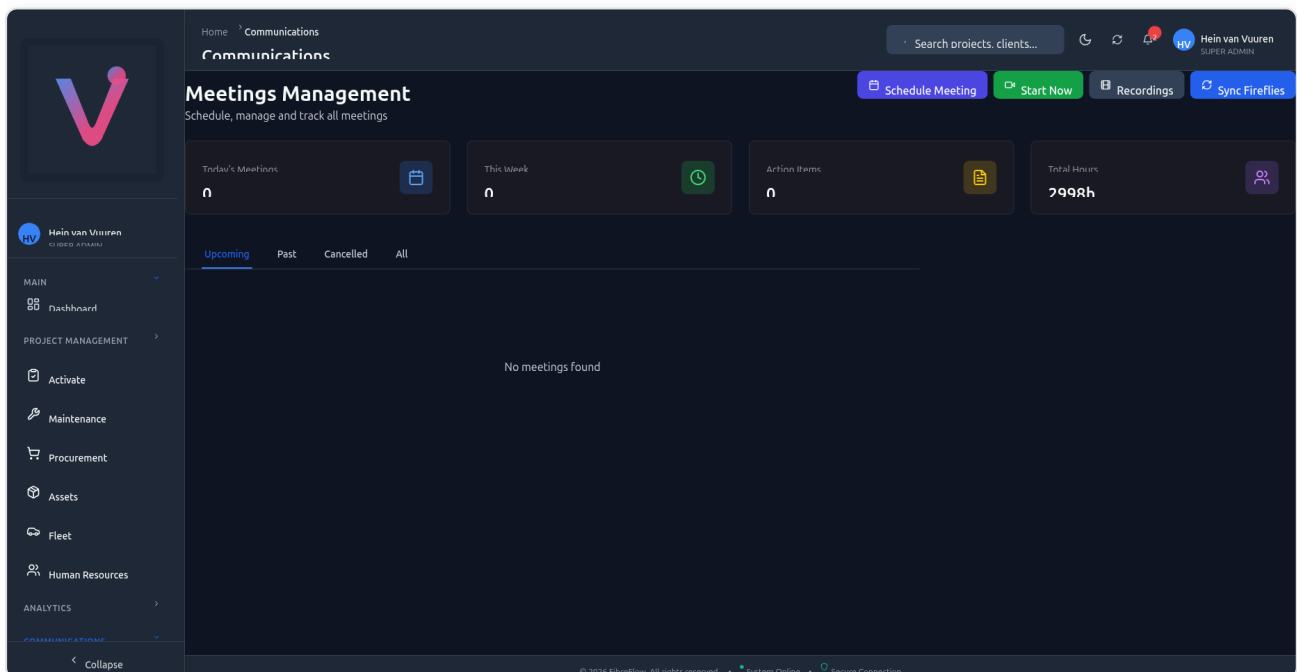


Figure 2.2: Meetings page showing synced transcripts with search and date filters

FEATURES

- **Auto-sync** — Meetings from Fireflies.ai are automatically synced to FibreFlow

- **Manual sync** — Click **Sync Now** to pull the latest transcripts
- **Meeting detail** — Click any meeting to view:
 - Full transcript
 - AI-generated summary with key points
 - Action items extracted from the meeting
 - Participants list
 - Duration and date

MEETING DETAIL MODAL

Section	Content
Summary	AI-generated meeting summary
Keywords	Key topics discussed
Action Items	Tasks identified from the conversation
Outline	Structured breakdown of the meeting
Participants	List of attendees

2.3 Action Items

Track tasks and follow-ups that come out of meetings, projects, or ad-hoc assignments.

Navigation: Sidebar → **Main** → **Action Items**

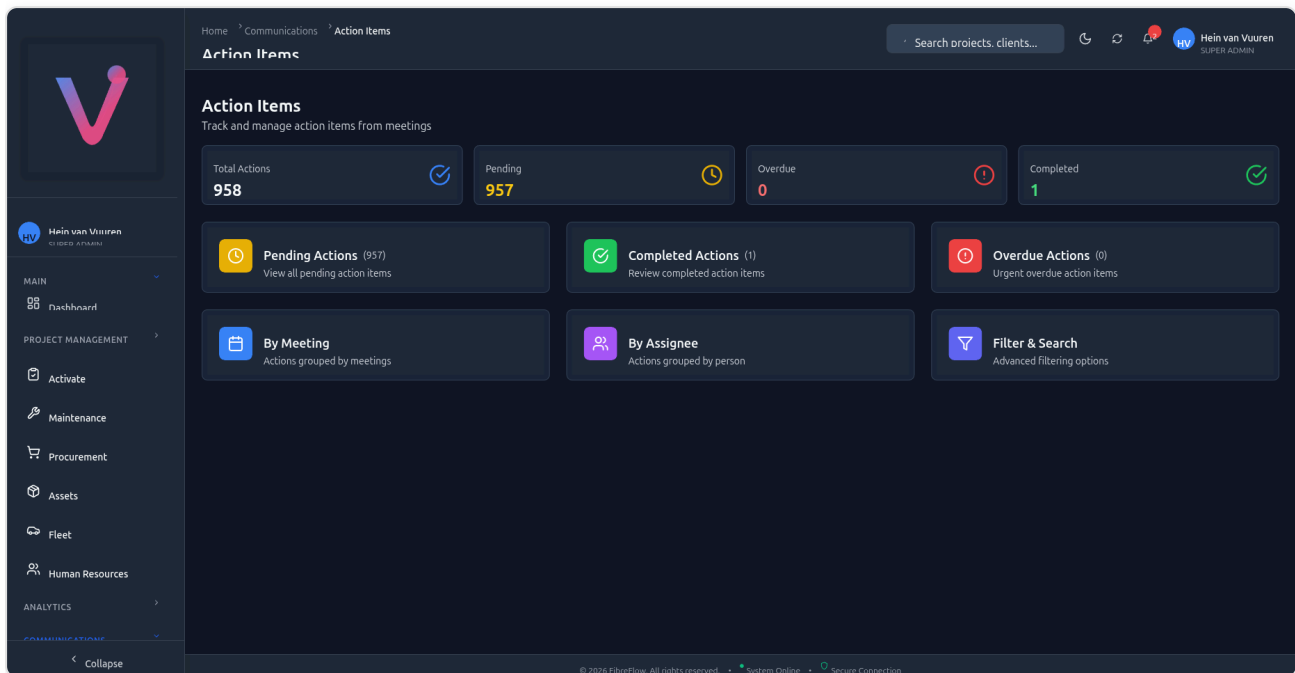


Figure 2.3: Action Items list with status filters and assignee tracking

FEATURES

- **Create** — Add new action items with title, description, assignee, and due date
- **Filter** — Filter by status (Open, In Progress, Completed) or assignee
- **Track** — Monitor progress with status updates and due date tracking
- **Link** — Action items can be linked to meetings or projects

3. Project Management

3.1 Projects List

The Projects module is the core of FibreFlow, managing fiber network deployment projects from planning through completion.

Navigation: Sidebar → **Project Management** → **Projects**

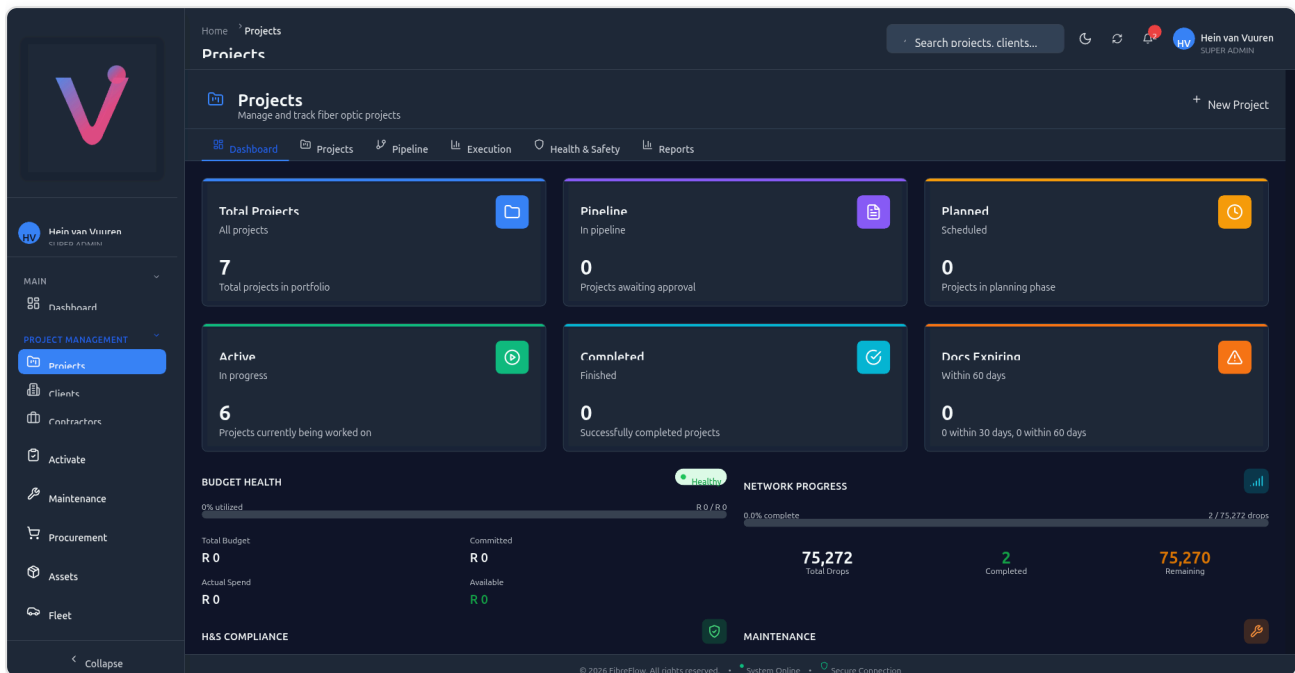


Figure 3.1: Projects list showing active projects with status, progress, and quick filters

VIEWS

- **Grid View** — Card-based layout showing project cards with key metrics
- **List View** — Table layout with sortable columns

FILTERS

Filter	Options
Status	All, Active, Planning, On Hold, Completed
Search	Search by project name, code, or client
Sort	Name, Date Created, Status, Progress

CREATING A NEW PROJECT

1. Click **+ New Project** (top-right)
2. Fill in the required fields:
 - **Project Name** — Descriptive name
 - **Client** — Select from client list
 - **Project Code** — Auto-generated or manual

- **Status** — Default: Planning
- **Start Date** and **Target End Date**

3. Click **Save**

3.2 Project Detail

Click any project to view its detail page with comprehensive tabs.

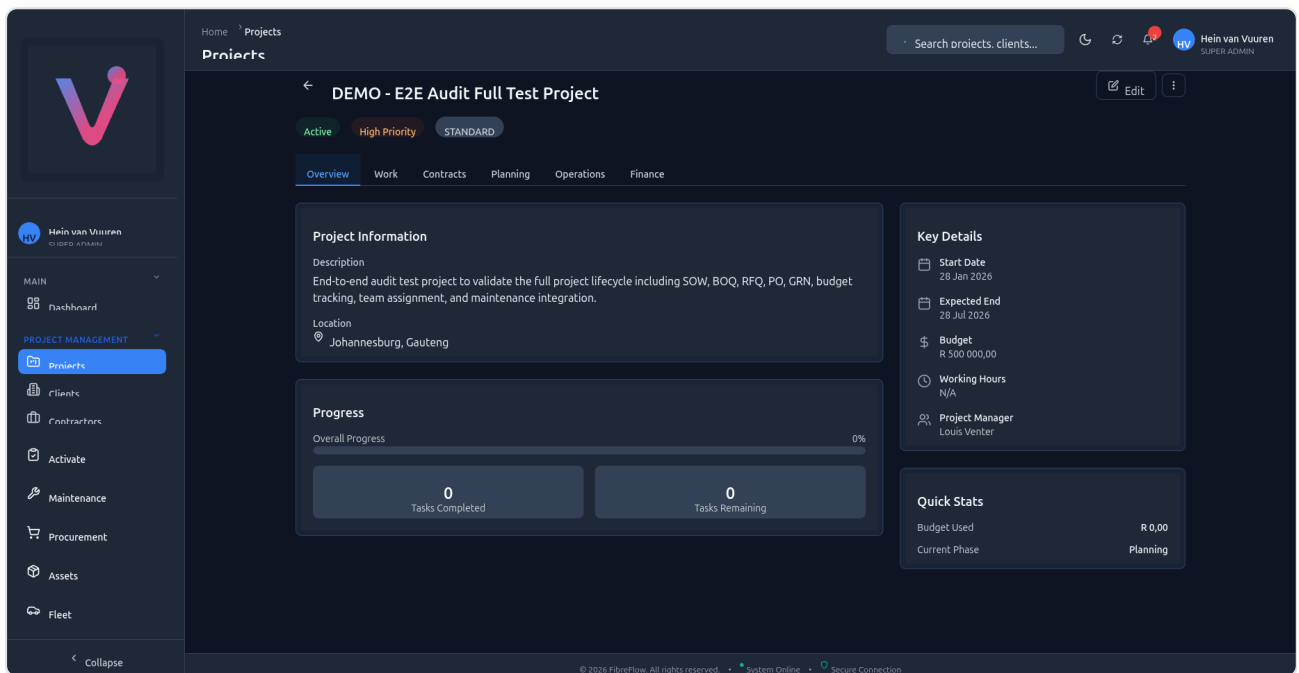


Figure 3.2: Project detail page showing overview tab with team, timeline, and key metrics

TABS

Tab	Content
Overview	Project info, status, progress, key metrics
Team	Assigned staff and contractors with roles
Timeline	Activity timeline and milestones
Procurement	BOQ, RFQ, PO summary for this project
Health & Safety	H&S compliance, incidents, checklists
Reports	Project-specific reports

TEAM MANAGEMENT

The Team tab shows a unified view of all team members (staff and contractors):

- **Primary Manager** — The project lead (marked with star icon)
- **Team Members** — Staff assigned to this project
- **Contractors** — External contractors working on the project
- Add/remove members using the **+ Add Member** button

3.3 Pipeline

The Pipeline module tracks project progression through pre-construction phases — from initial proposal to construction readiness.

Navigation: Sidebar → **Project Management** → **Projects** → **Pipeline** tab

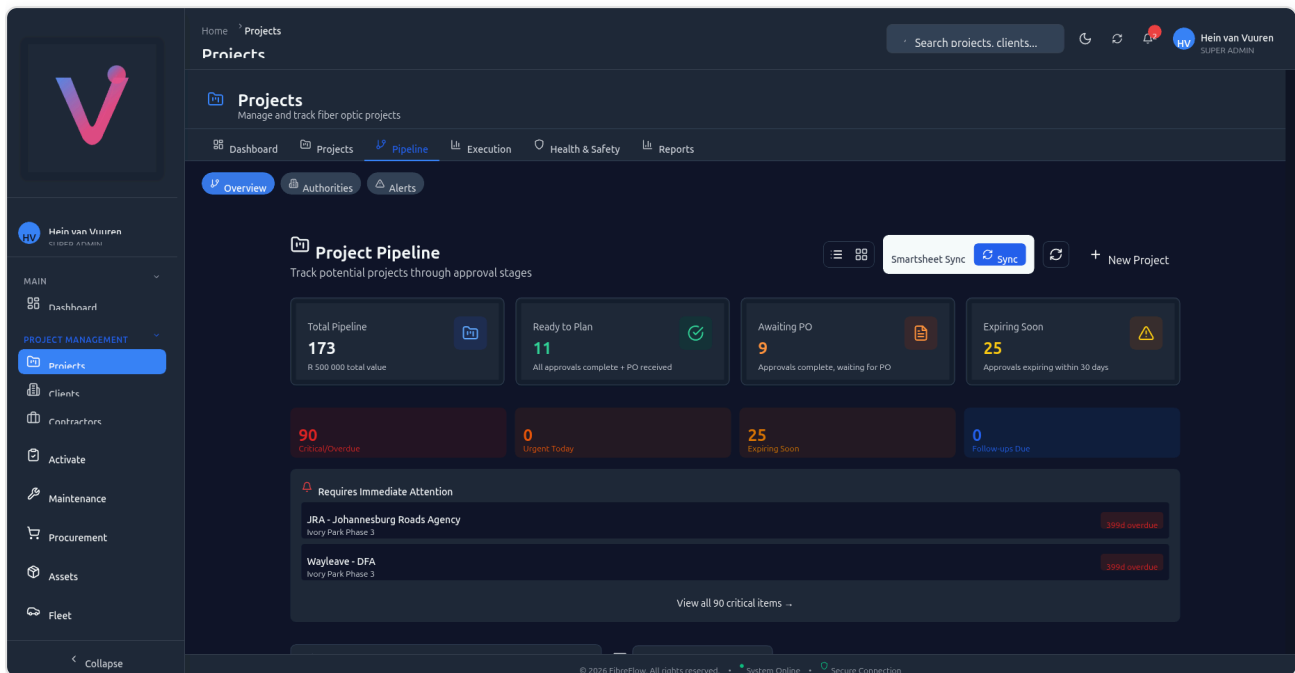


Figure 3.3: Pipeline view showing projects across approval stages

SUB-TABS

Tab	Purpose
Overview	Pipeline stages and project positioning
Authorities	Service authority management and approvals
Alerts	Pipeline alerts and blockers

PIPELINE STAGES

Projects move through defined stages:

1. **Proposal** — Initial project proposal
2. **Due Diligence** — Feasibility study and site surveys
3. **Approvals** — Municipal and authority approvals
4. **Design** — Network design and engineering
5. **Ready for Construction** — All approvals received

AUTHORITIES

Manage service authorities required for each project:

- Add authority requirements (municipal, environmental, utility)
- Track submission and approval status
- Upload supporting documents
- Set expiry dates

3.4 Execution & Progress

Track daily progress and manage project tasks during the construction phase.

Navigation: Sidebar → **Project Management** → **Projects** → **Execution** tab

PROGRESS REPORTS

- View daily progress entries by project
- Track meters of fiber installed, poles erected, drops connected
- Compare planned vs actual progress

TASKS

- Create and assign project tasks
- Set priorities and due dates
- Track task completion status

3.5 Health & Safety

Manage health and safety compliance for each project.

Navigation: Sidebar → **Project Management** → **Projects** → **Health & Safety** tab

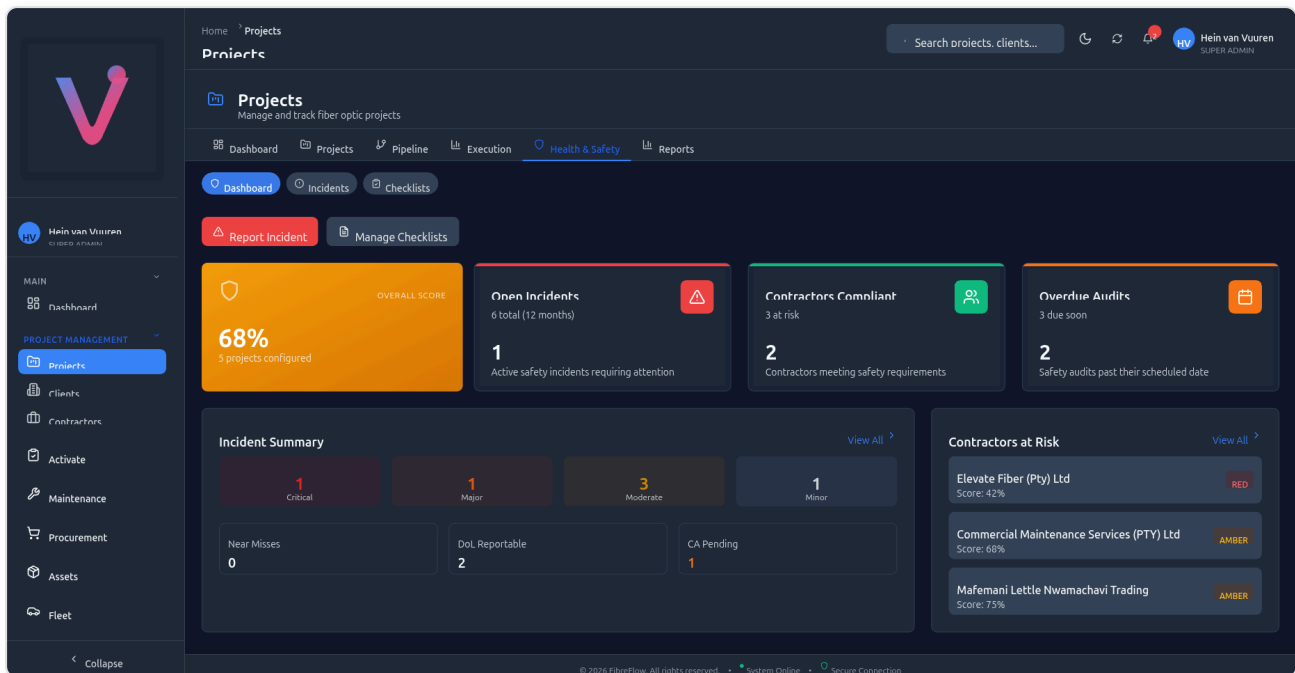


Figure 3.4: Health & Safety dashboard showing incident tracking and compliance scores

SUB-TABS

Tab	Purpose
Dashboard	H&S compliance overview and statistics
Incidents	Record and track safety incidents
Checklists	Safety inspection checklists

INCIDENT MANAGEMENT

- Record incidents with type, severity, date, and description
- Attach photos and documents
- Track investigation status and corrective actions
- View incident trends over time

3.6 Clients

Manage client organizations associated with your fiber projects.

Navigation: Sidebar → **Project Management** → **Clients**

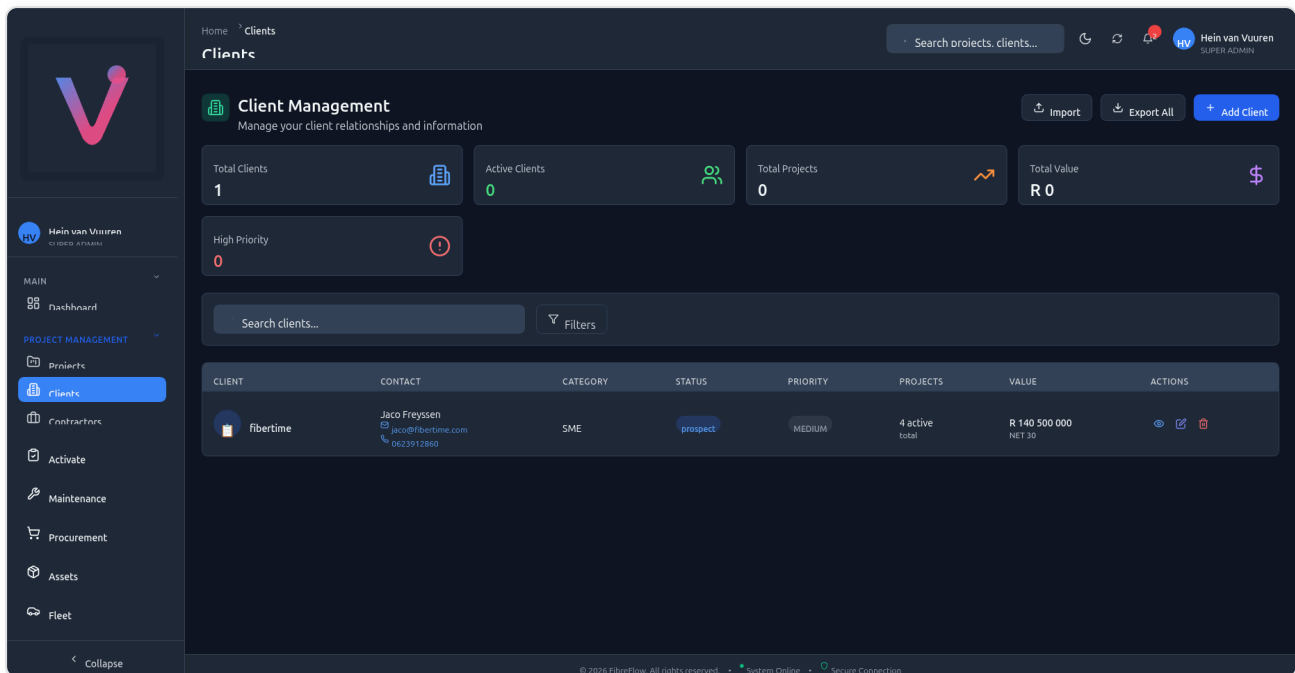


Figure 3.5: Clients list showing organization details and project counts

FEATURES

- **Client List** — View all clients with search and status filters
- **Client Detail** — Click to view full details including:
 - Contact information (name, phone, email, address)
 - Associated projects
 - Financial summary (total revenue)
 - Notes and communications history
- **Create Client** — Add new client organizations
- **Edit/Delete** — Manage existing client records

3.7 Contractors

Manage external contractors and their performance.

Navigation: Sidebar → **Project Management** → **Contractors**

FEATURES

- **Contractor Portal** — List of all contractors with project assignments
- **RAG Dashboard** — Red/Amber/Green performance ratings:

- **Red** — Significant issues, action required
 - **Amber** — Minor issues, monitor closely
 - **Green** — Performing well
- **Document Reports** — Track contractor document compliance (insurance, licenses, etc.)

3.8 Reports

Project-level reporting and analytics.

Navigation: Sidebar → **Project Management** → **Projects** → **Reports** tab

- Project progress reports
- Timeline adherence tracking
- Budget vs actual spending
- Resource utilization reports

4. Activate — QA & Photo Review

The Activate module is FibreFlow's AI-powered quality assurance system for verifying field installations through photo review.

4.1 Dashboard

Navigation: Sidebar → **Activate** (click section header)

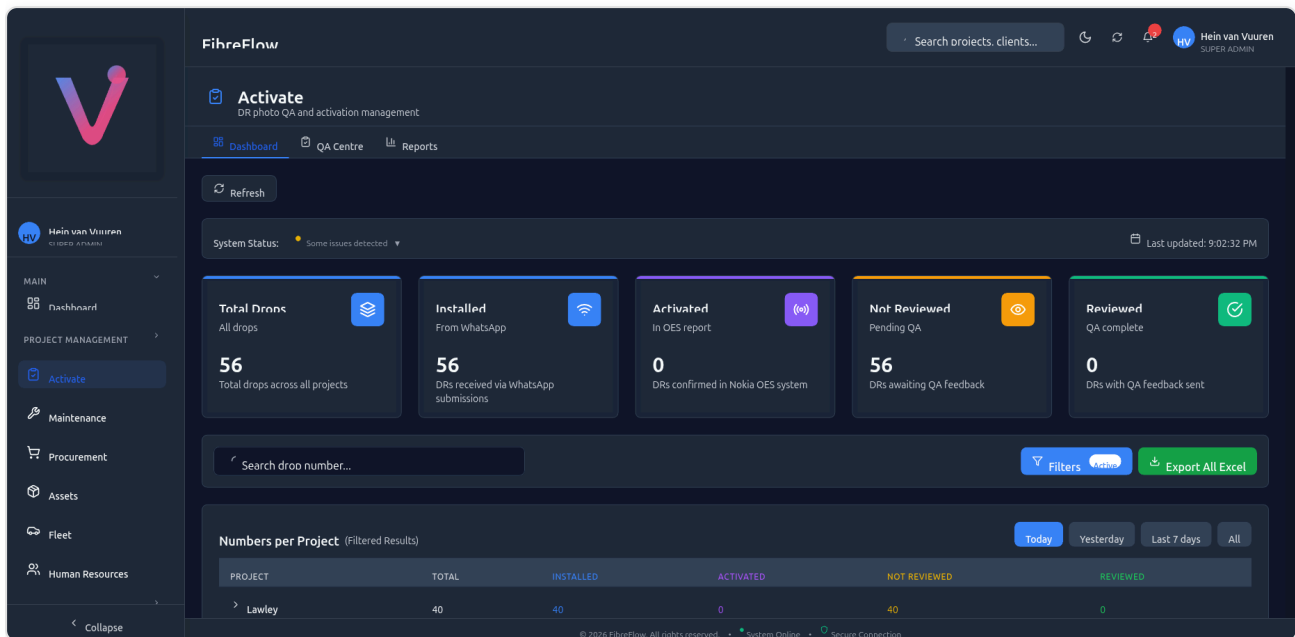


Figure 4.1: Activate Dashboard showing QA statistics, project filter, and drop counts by status

STATISTICS CARDS

Card	Description
Total DRs	Total drop reference numbers in the system
Pending QA	DRs awaiting quality review
Passed	DRs that passed QA review
Failed	DRs that failed QA review
Rework	DRs sent back for corrections

PROJECT FILTER

Select a specific project to filter the dashboard data, or view across all projects.

4.2 QA Centre

The QA Centre is the main workspace for reviewing field installation photos.

Navigation: Sidebar → **Activate** → **QA Centre** tab

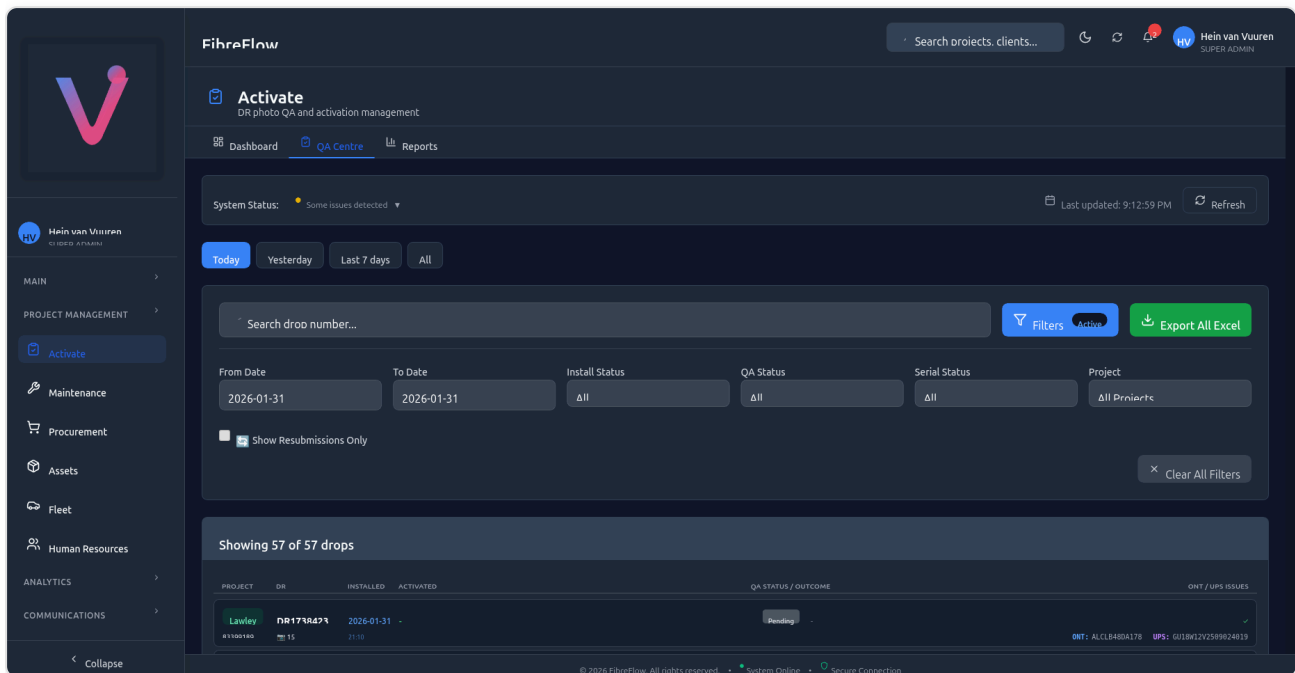


Figure 4.2: QA Centre showing DR list with status badges, search, and filter options

DR LIST

A searchable, filterable list of all drop reference numbers (DRs):

Column	Description
DR Number	Unique drop reference (e.g., DR1738371)
Project	Associated project name
Status	QA status badge (Pending, Passed, Failed, Rework)
Photos	Number of photos available
Submitted	Date the DR was submitted for review
Reviewed	Date of last QA review

SEARCH & FILTERS

- **Search** — Find DRs by number, project, or installer name
- **Status Filter** — Pending, Passed, Failed, Rework, All
- **Project Filter** — Filter by specific project

- **Date Range** — Filter by submission date

4.3 QA Wizard — 5-Phase Review

Click any DR to open the QA Wizard — a guided 5-phase review process.

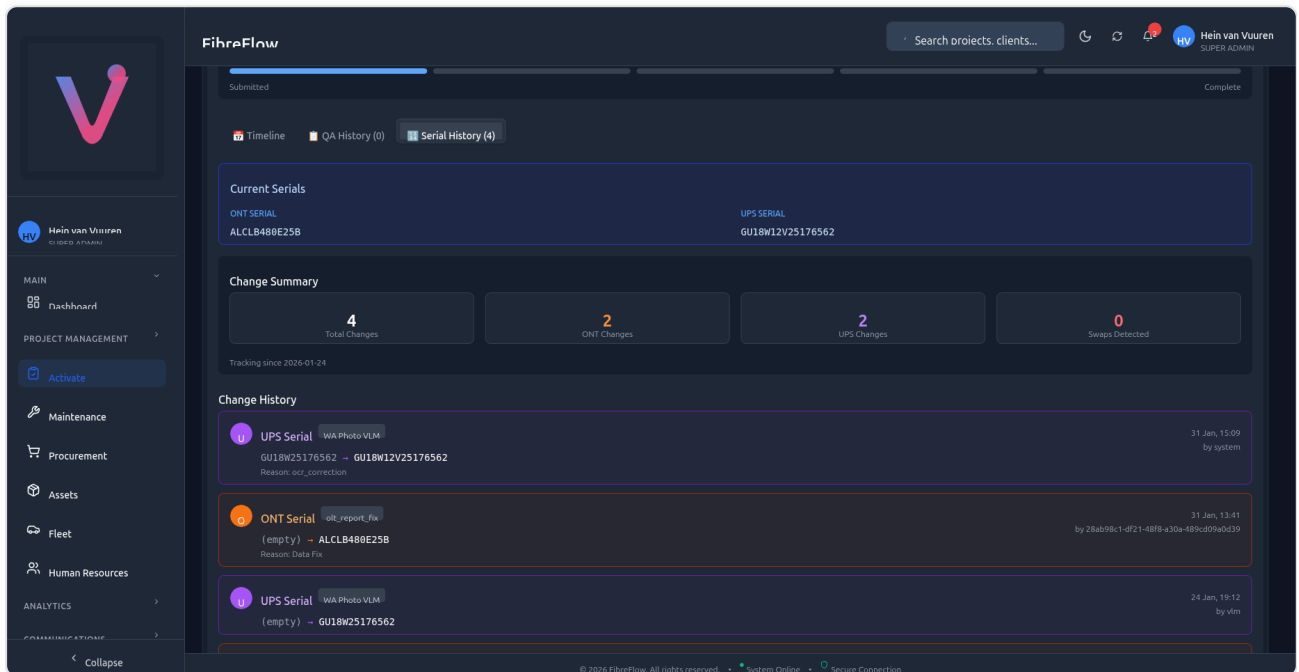


Figure 4.3: QA Wizard showing the photo review phase with categorized installation photos

PHASE 1: PREREQUISITES

Validates that the DR has sufficient data for review:

- Minimum photo count met
- Photos have been AI-categorized
- OES data available
- Contact information present

PHASE 2: PHOTO REVIEW

Review installation photos mapped to a 10-step checklist:

Step	Label	What to Check
1	House Photo	Property correctly identified
2	Cable from Pole	Cable routed properly from pole
3	Entry Outside	External cable entry point
4	Entry Inside	Internal cable entry point
5	Wall	Wall mount installation
6	ONT Back	ONT device installed correctly
7	Power Meter	Signal strength reading (-18 to -24 dBm)
8	Final Installation	Completed installation overview
9	Green Lights	ONT LED indicators showing green
10	Signature	Customer sign-off obtained

For each step:

- View the assigned photo(s)
- Approve or flag issues
- Re-assign photos to different steps if AI categorization was incorrect

PHASE 3: DATA VALIDATION

Validates extracted data:

- **Power meter reading** — Must be between -18 and -24 dBm
- **ONT serial** — Extracted serial matches expected format (ALCL prefix)
- **DR number** — Matches the submission

PHASE 4: FINAL DECISION

Make the QA decision:

Decision	Action	Result
Pass	Installation meets all requirements	DR marked as approved
Fail	Critical issues found	DR marked as failed with reason codes
Rework Needed	Minor issues to fix	DR sent back to installer with feedback

Select a reason code when failing or requesting rework:

- Missing photos
- Poor photo quality
- Installation defect
- Serial mismatch
- Signal out of range

PHASE 5: FEEDBACK

Generate and send feedback to the field installer:

- Auto-generated feedback message based on decision and reason codes
- Sent via WhatsApp to the installer
- Includes specific issues to address
- Tracks acknowledgment status

4.4 VLM AI Integration

FibreFlow uses a Vision Language Model (VLM) — Qwen3-VL — to automatically process installation photos:

- **Auto-categorization** — Photos are categorized to the 10-step checklist
- **Serial Extraction** — ONT serial numbers are read from photos
- **Power Reading** — dBm values extracted from power meter photos
- **DR Detection** — DR numbers identified from labels

***Tip:** AI categorization is a starting point — always verify and correct during the Photo Review phase. Your corrections are saved and improve future AI accuracy through few-shot learning.*

4.5 Reports

Navigation: Sidebar → **Activate** → **Reports** tab

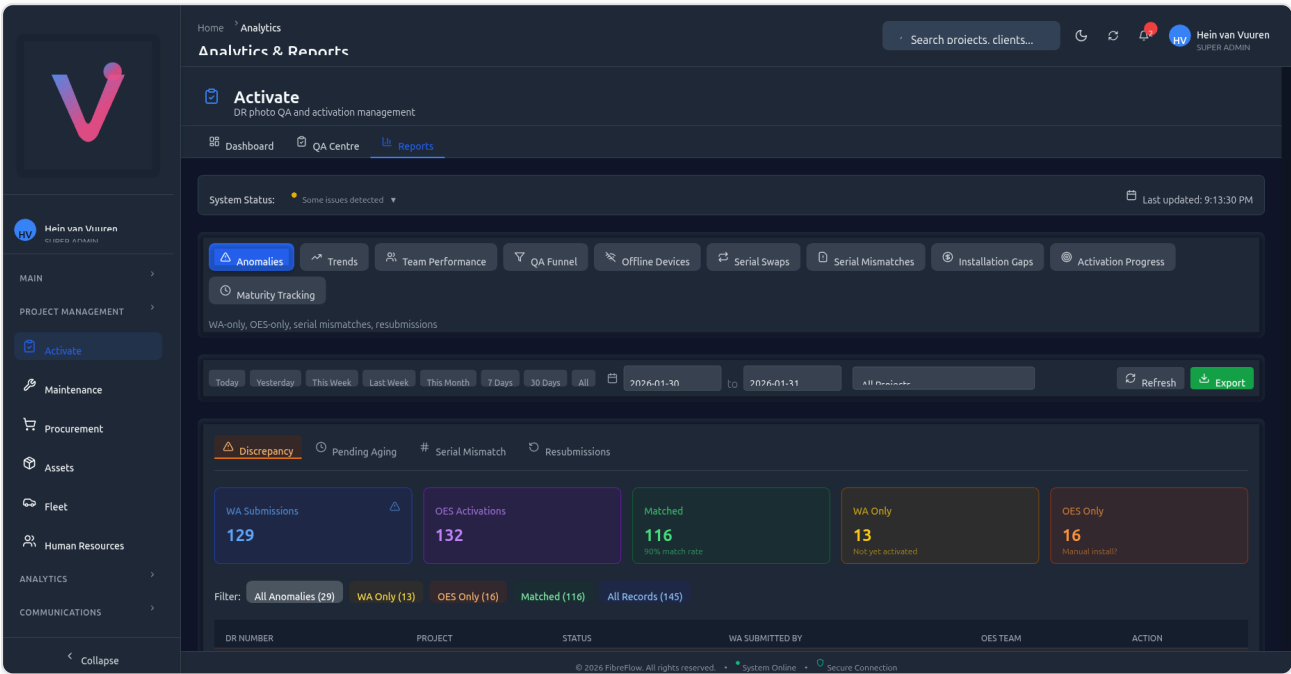


Figure 4.4: Activate Reports dashboard showing trend analysis and performance metrics

Available report types:

Report	Description
Trends	QA pass/fail rates over time
Funnel	DR progression through QA stages
Team Performance	Reviewer throughput and accuracy
Serial Validation	ONT serial match rates
Offline Devices	Installations with connectivity issues
Serial Mismatch	DRs where serial numbers do not match expected
Installation Gaps	Missing checklist steps
Anomaly Detection	Unusual patterns in submissions

All reports support:

- Date range filtering
- Project filtering
- CSV export

4.6 OES Import & Data Sync

(Admin only)

- **OES Import** — Import OES (Operation Engineering Support) activation data from Excel files
- **SharePoint Sync** — Synchronize DR data from SharePoint document libraries
- **QField Sync** — Push OES coordinate data to QField for field mapping

5. Maintenance

The Maintenance module manages the complete lifecycle of fiber network maintenance tickets.

Note: A comprehensive standalone Maintenance manual is available with detailed coverage of all features. This chapter provides an overview — see the dedicated **FibreFlow Maintenance Module User Manual** for in-depth documentation.

5.1 Dashboard

Navigation: Sidebar → **Maintenance** (click section header)

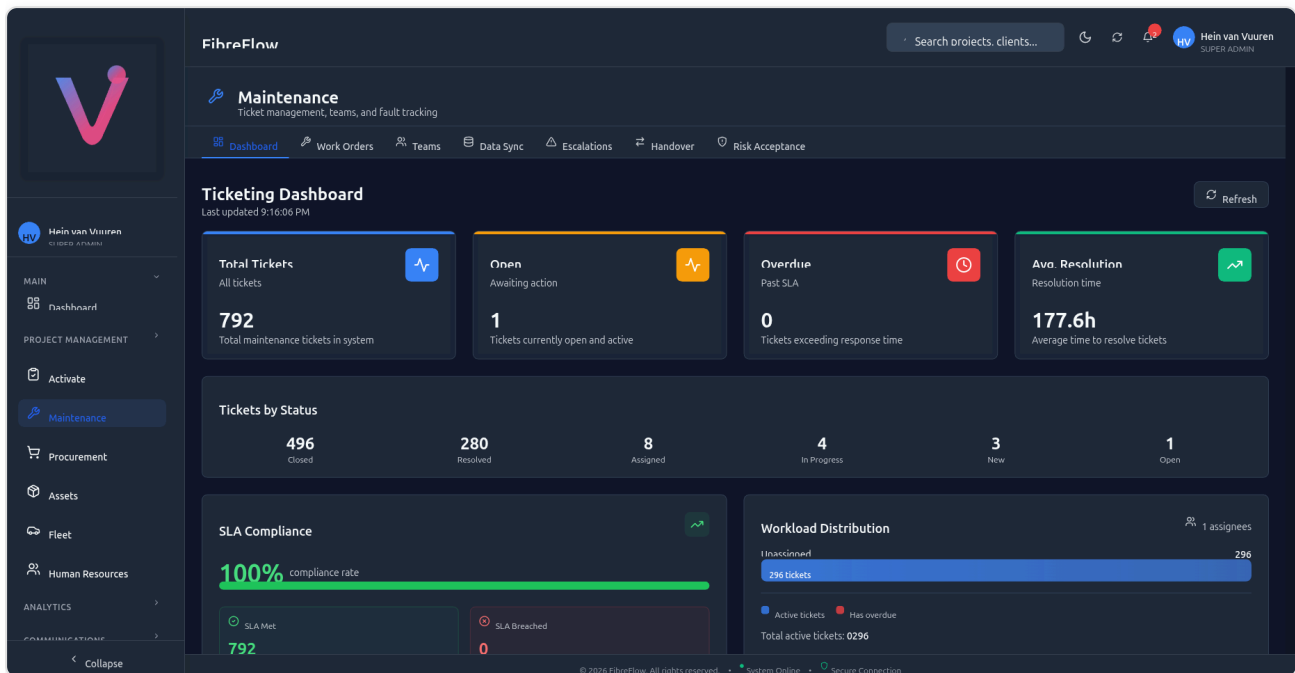


Figure 5.1: Maintenance Dashboard showing ticket statistics, SLA compliance, and workload

KEY METRICS

Metric	Description
Total Tickets	All maintenance tickets
Open	Tickets awaiting action
Overdue	Tickets exceeding SLA target
Avg. Resolution	Average resolution time
SLA Compliance	Percentage meeting response targets

5.2 Work Orders — Kanban Board

The primary ticket management view with drag-and-drop columns.

Navigation: Sidebar → **Maintenance** → **Work Orders** tab

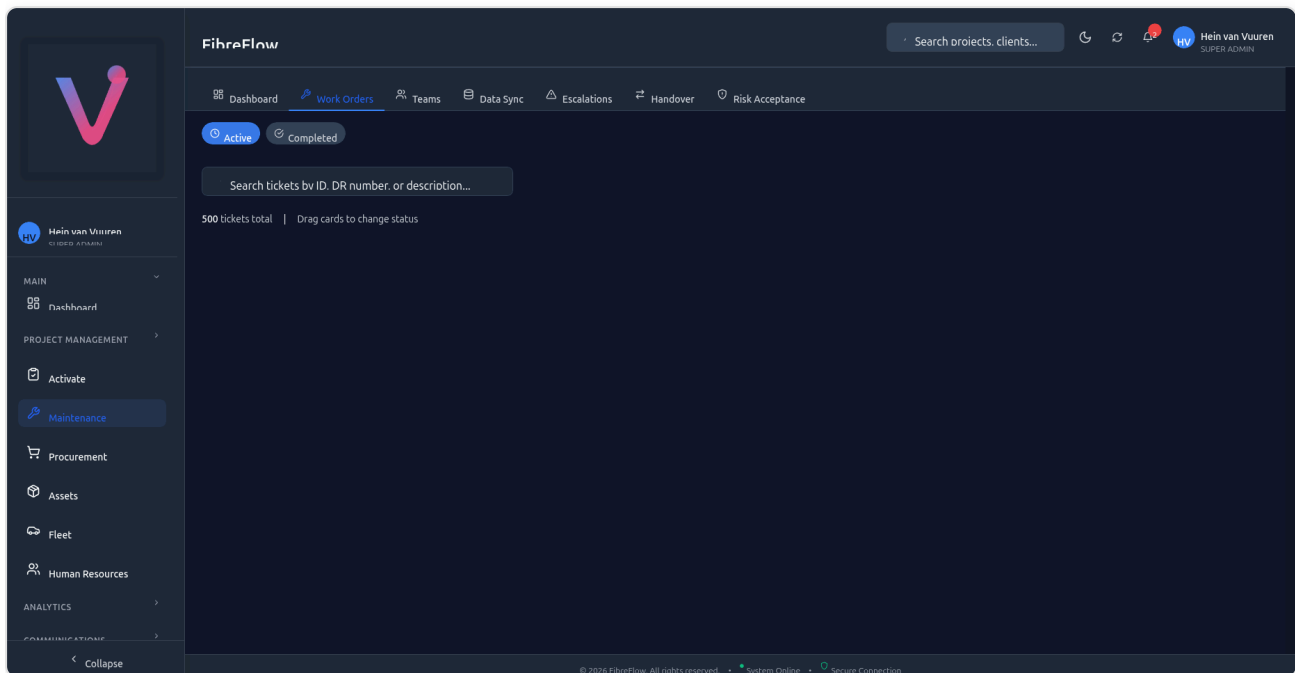
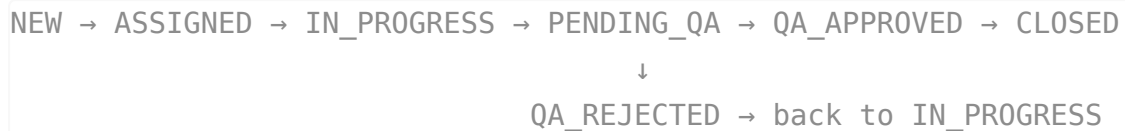


Figure 5.2: Maintenance Kanban Board with tickets organized by status columns

COLUMNS

Column	Description
New	Newly created tickets
Triaged	Assessed and categorized
Assigned	Assigned to a technician
In Progress	Work actively underway
Blocked	Blocked by an external issue
Resolved	Work completed
Closed	Ticket fully closed

TICKET LIFECYCLE



5.3 Key Features Summary

Feature	Description	See Standalone Manual Section
Ticket Creation	Create tickets manually or via import	Section 6
Team Management	Organize internal and contractor teams	Section 8
QContact Sync	Bidirectional sync with FiberTime QContact CRM	Section 9
Escalations	Automatic repeat fault detection	Section 10
Handover	5-gate controlled team transitions	Section 11
Risk Acceptance	Conditional QA approvals	Section 12

5.4 SLA Targets

Priority	SLA Target
Critical	6 hours
Urgent	12 hours
High	1 day
Normal	3 days
Low	7 days

5.5 Fault Cause Attribution

Every maintenance ticket requires a fault cause — this determines contractor liability:

Fault Cause	Contractor Liable?
Workmanship	YES
Material Failure	No
Client Damage	No
Third Party	No
Environmental	No
Vandalism	No
Unknown	No

Important: Fault cause selection directly affects contractor billing. Choose accurately.

6. Procurement

The Procurement module manages the complete purchasing workflow — from Bill of Quantities through to goods receipt.

6.1 Dashboard

Navigation: Sidebar → **Procurement** (click section header)

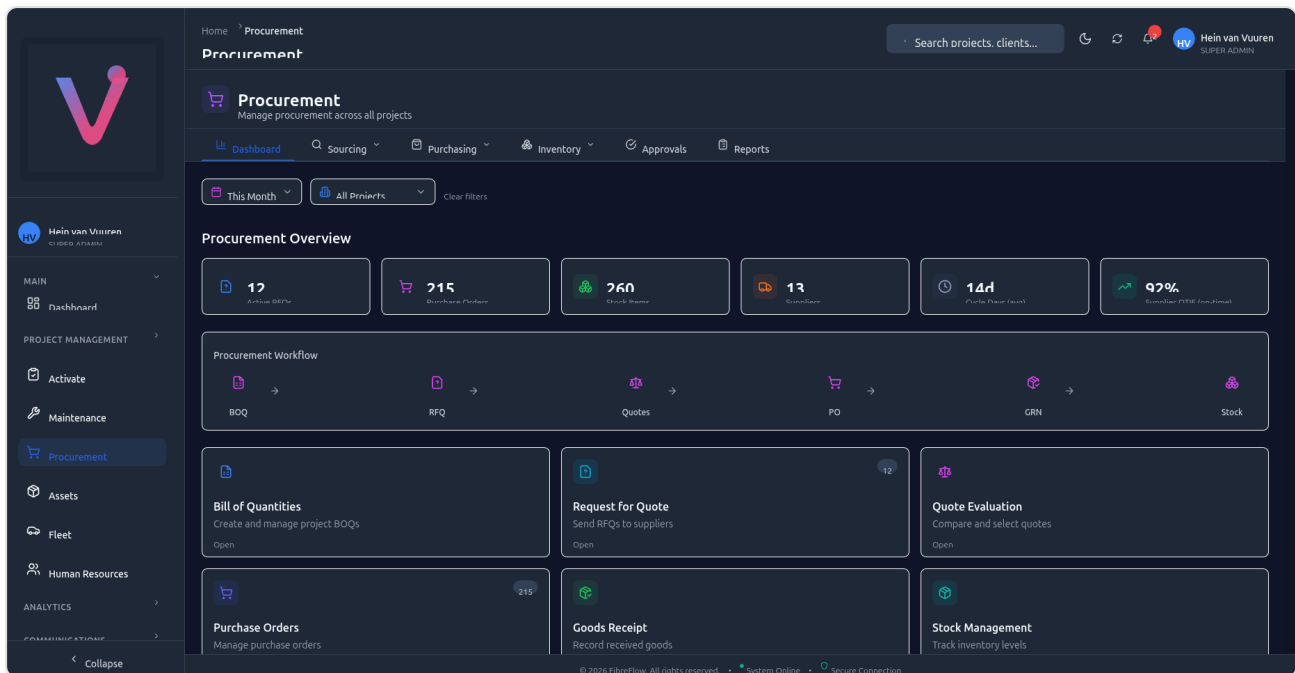


Figure 6.1: Procurement Dashboard showing order statistics, pending approvals, and quick actions

QUICK ACTIONS

- **Create BOQ** — Start a new Bill of Quantities
- **New RFQ** — Create a Request for Quotation
- **Create PO** — Generate a Purchase Order
- **View Approvals** — Check pending approval items

DASHBOARD TABS

The procurement module uses horizontal tabs for navigation:

Tab	Purpose
Dashboard	Overview statistics and quick actions
Sourcing	BOQ management
Purchasing	RFQ and PO management
Inventory	Stock items, movements, stock takes
Financial	Cost tracking and budget overview
Approvals	Pending approval items

6.2 Sourcing — BOQ Management

The Bill of Quantities (BOQ) manages material requirements for projects.

Navigation: Procurement → **Sourcing** tab

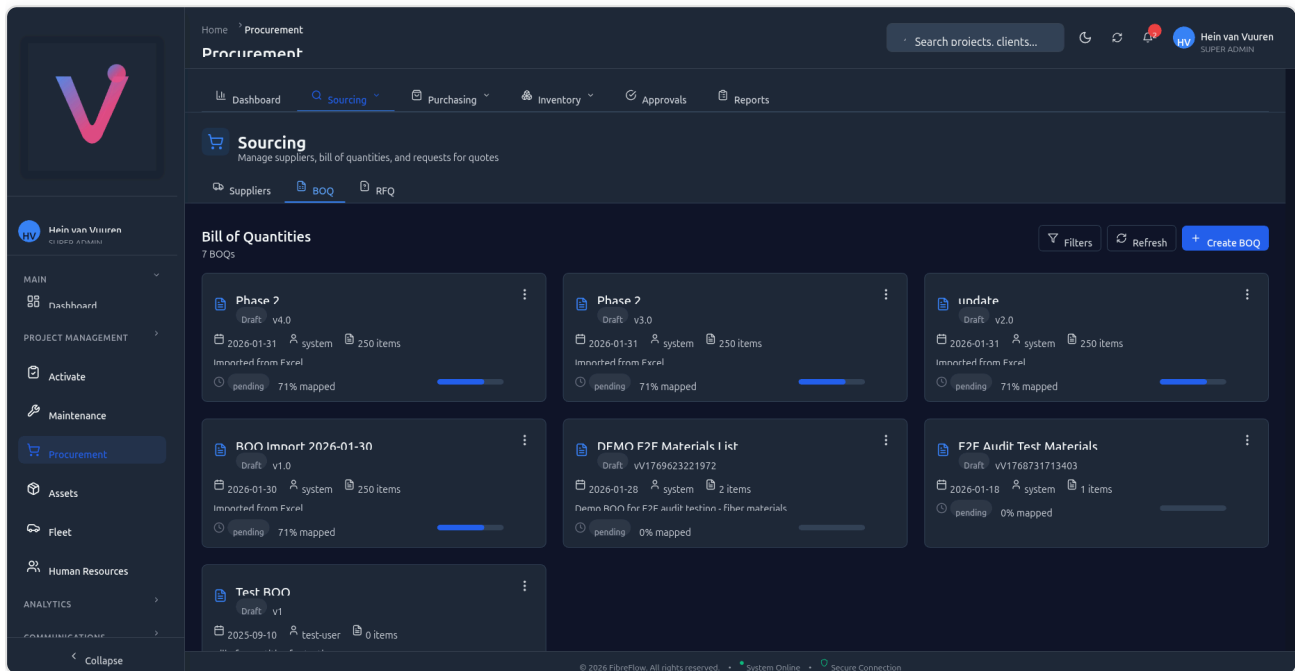


Figure 6.2: BOQ management showing project BOQs with upload and column mapping features

CREATING A BOQ

1. Click **+ Create BOQ**
2. Select the **Project**
3. **Upload Excel file** — The system auto-detects columns:
 - Item description
 - Quantity
 - Unit of measure
 - Unit price
4. **Column Mapping** — Review and adjust detected columns
5. **Stock Matching** — System matches BOQ items to existing stock items using fuzzy matching
6. Review and **Confirm Import**

BOQ FEATURES

- **Upload** — Import from Excel with auto-detection
- **Column Mapper** — Visual interface to map Excel columns to BOQ fields
- **Stock Matcher** — Fuzzy matching to link BOQ items to inventory
- **History** — Track BOQ revisions and changes
- **Export** — Download BOQ as Excel

6.3 RFQ — Request for Quotation

Send requests for quotation to suppliers and compare responses.

Navigation: Procurement → **Purchasing** tab → RFQ section

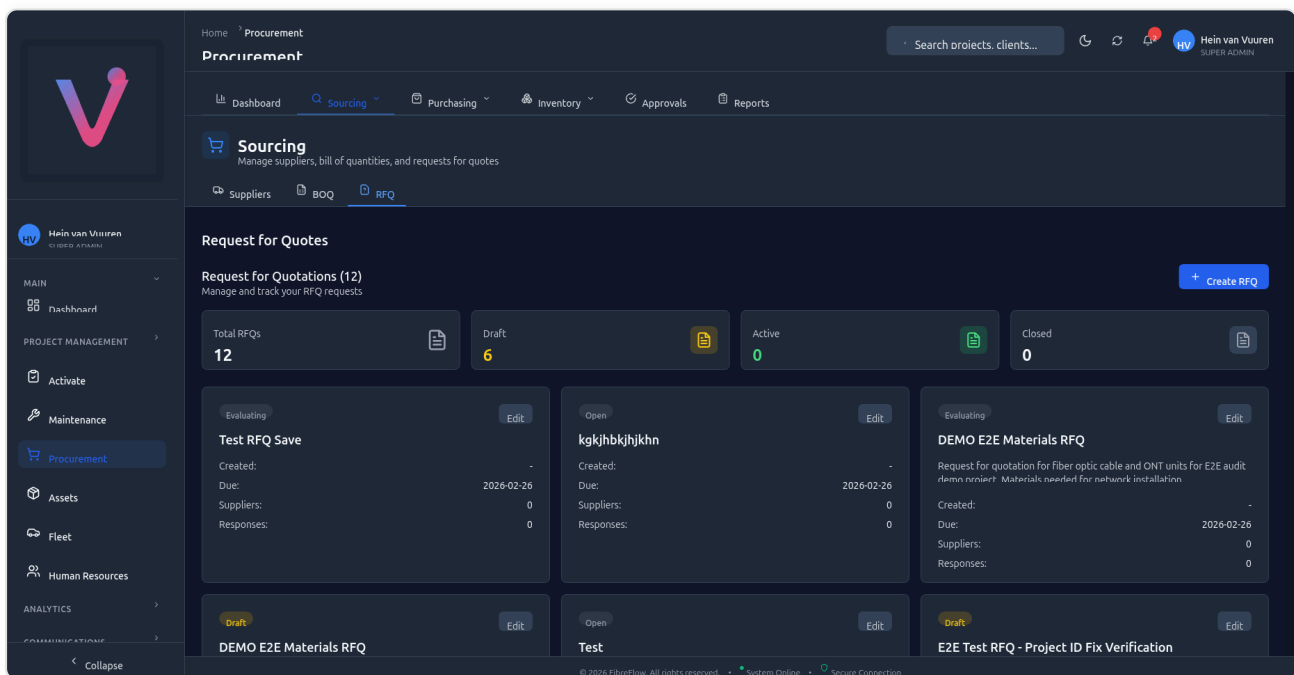


Figure 6.3: RFQ detail page showing line items, supplier responses, and comparison view

CREATING AN RFQ

1. Navigate to **Purchasing** tab
2. Click **+ New RFQ**
3. Fill in:
 - **Title** — RFQ description
 - **Project** — Associated project

- **Due Date** — Response deadline
- **Items** — Add line items (from BOQ or manual entry)

4. Select **Suppliers** to send to

5. Click **Create & Send**

MANAGING RFQ RESPONSES

- View responses from each supplier
- Compare prices side-by-side
- Select winning quotes
- **Convert to PO** — Create a purchase order from the selected response

6.4 Purchase Orders

Create and manage purchase orders with approval workflows.

Navigation: Procurement → **Purchasing** tab → PO section

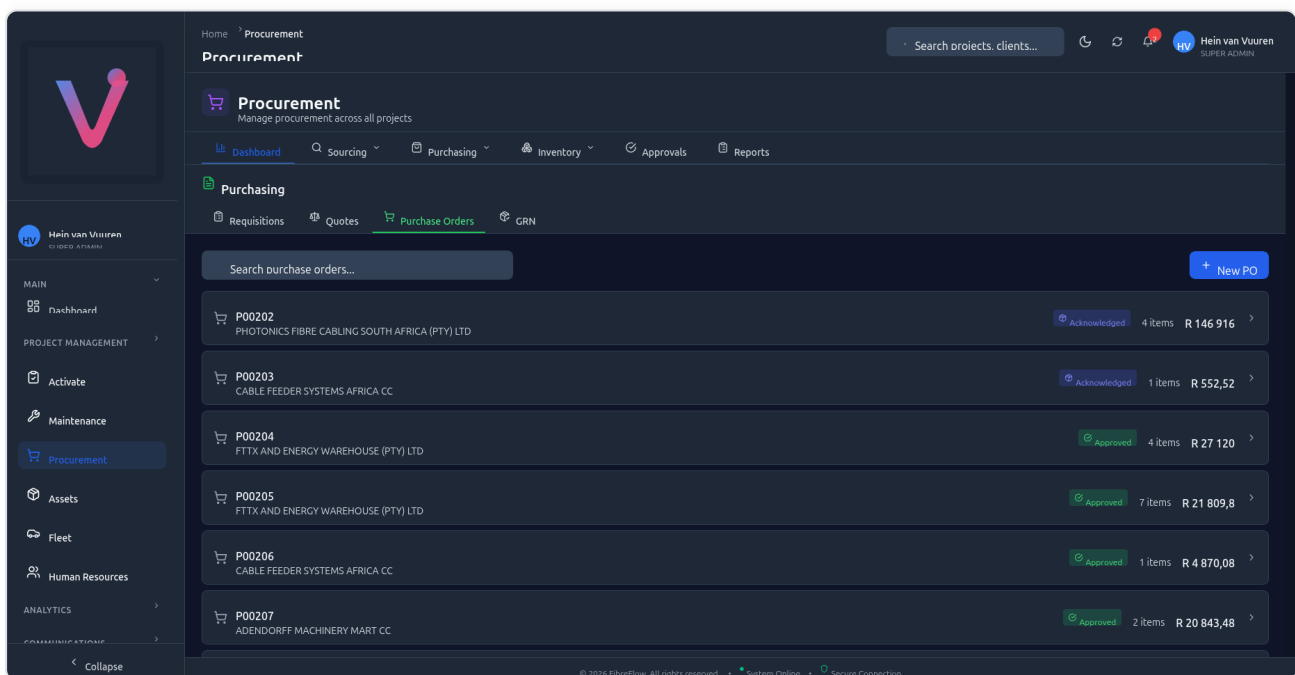


Figure 6.4: Purchase Order detail showing line items, approval status, and version history

PO APPROVAL WORKFLOW

Purchase orders follow an approval chain based on value thresholds:

Value Range	Required Approval
R0 – R10,000	Manager approval
R10,001 – R50,000	Senior manager + Director
R50,001+	Executive approval

PO STATUSES

Status	Description
Draft	PO being prepared
Pending Approval	Awaiting approver sign-off
Approved	Ready to send to supplier
Sent	Sent to supplier
Partially Received	Some items received
Fully Received	All items received
Cancelled	PO cancelled

PO VERSIONING

When a PO is modified after approval, a new version is created. The version history tracks all changes with timestamps and reasons.

6.5 Inventory

Manage stock items, track movements, and perform stock takes.

Navigation: Procurement → **Inventory** tab

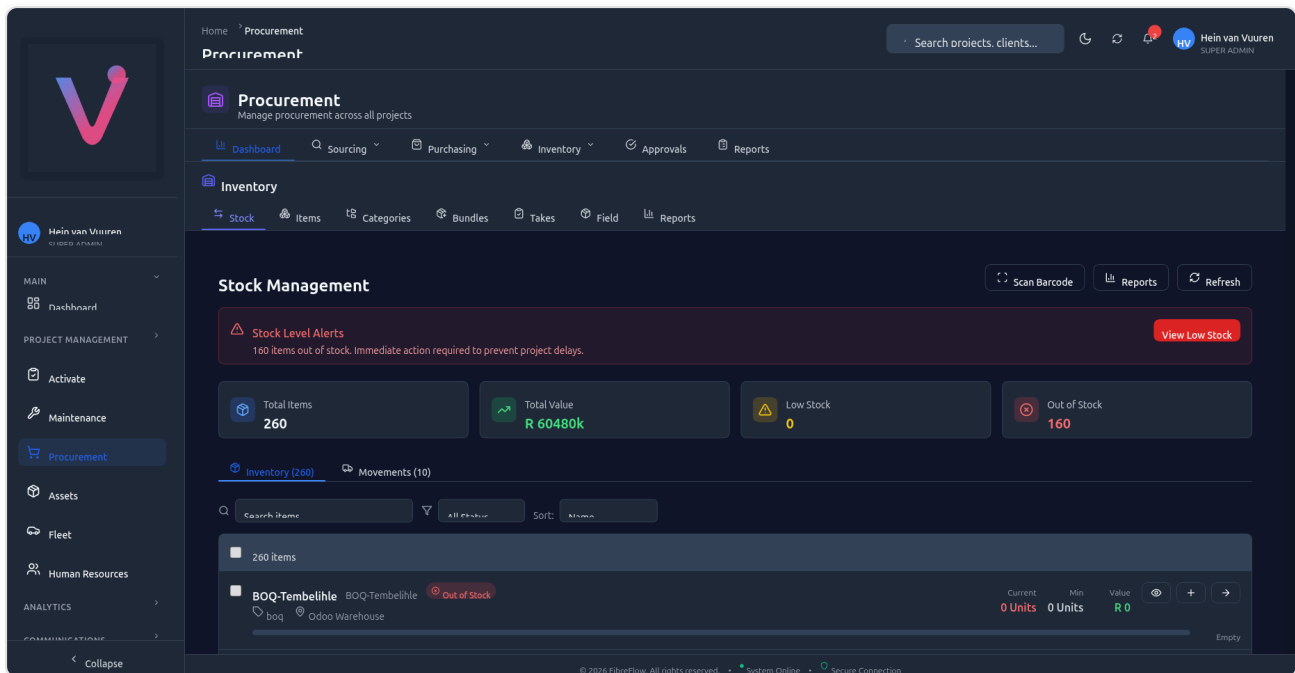


Figure 6.5: Stock management dashboard showing items, levels, and movement history

STOCK ITEMS

- View all stock items with search and category filters
- Visual stock level bars showing quantity vs max capacity
- Quick filters for **Low Stock** and **Out of Stock**
- Color-coded status indicators:
 - **Green** — Adequate stock
 - **Yellow** — Low stock (below minimum)
 - **Red** — Out of stock

STOCK MOVEMENTS

Track all inventory movements:

- **Receipts** — Items received from suppliers (GRN)
- **Issues** — Items issued to projects
- **Transfers** — Items moved between locations
- **Adjustments** — Stock count corrections

GOODS RECEIPT NOTES (GRN)

Record received goods against purchase orders:

1. Navigate to a PO → Click **Receive Goods**
2. Enter quantities received for each line item
3. Note any discrepancies
4. Upload delivery note/photo
5. Confirm receipt

6.6 Financial

Cost tracking and budget overview for procurement activities.

Navigation: Procurement → **Financial** tab

- Total spend by project, supplier, and category
- Budget vs actual comparison
- Outstanding payment tracking
- Cost trends over time

6.7 Approvals

View and action pending approvals across procurement.

Navigation: Procurement → **Approvals** tab

Action	Description
Approve	Approve the purchase order
Reject	Reject with reason
Request Changes	Send back for modifications

6.8 Suppliers

Manage supplier relationships and item codes.

Navigation: (Accessed via Projects → Contractors or Procurement pages)

- **Supplier List** — All registered suppliers
- **Supplier Detail** — Contact info, item codes, performance

- **Item Codes** — Map supplier-specific codes to stock items
- **Performance** — On-time delivery, quality, pricing metrics

7. Assets

The Assets module provides complete asset lifecycle management for tools, equipment, and materials.

7.1 Dashboard

Navigation: Sidebar → **Assets** (click section header)

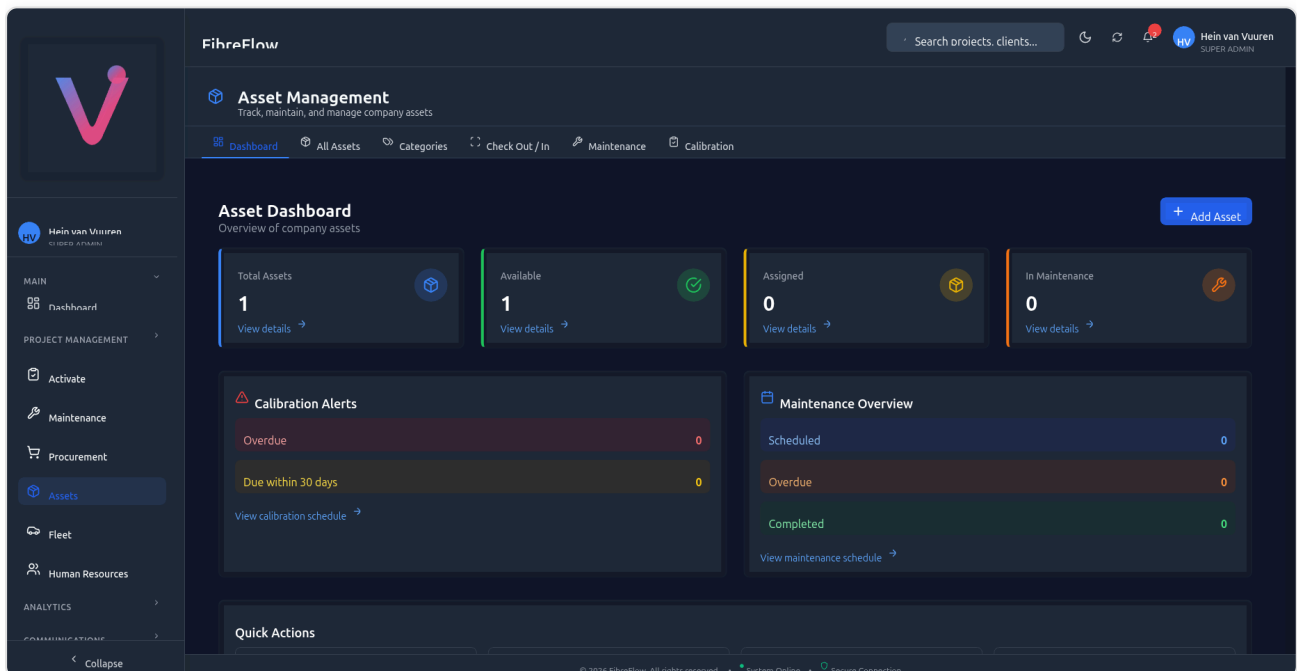


Figure 7.1: Assets Dashboard showing total assets, available, assigned, and maintenance alerts

STATISTICS CARDS

Card	Description
Total Assets	Total registered assets
Available	Assets ready for checkout
Assigned	Assets currently checked out
In Maintenance	Assets under repair/service
Calibration Due	Assets needing calibration
Calibration Overdue	Assets past calibration date

7.2 All Assets

Navigation: Assets → **All Assets** tab

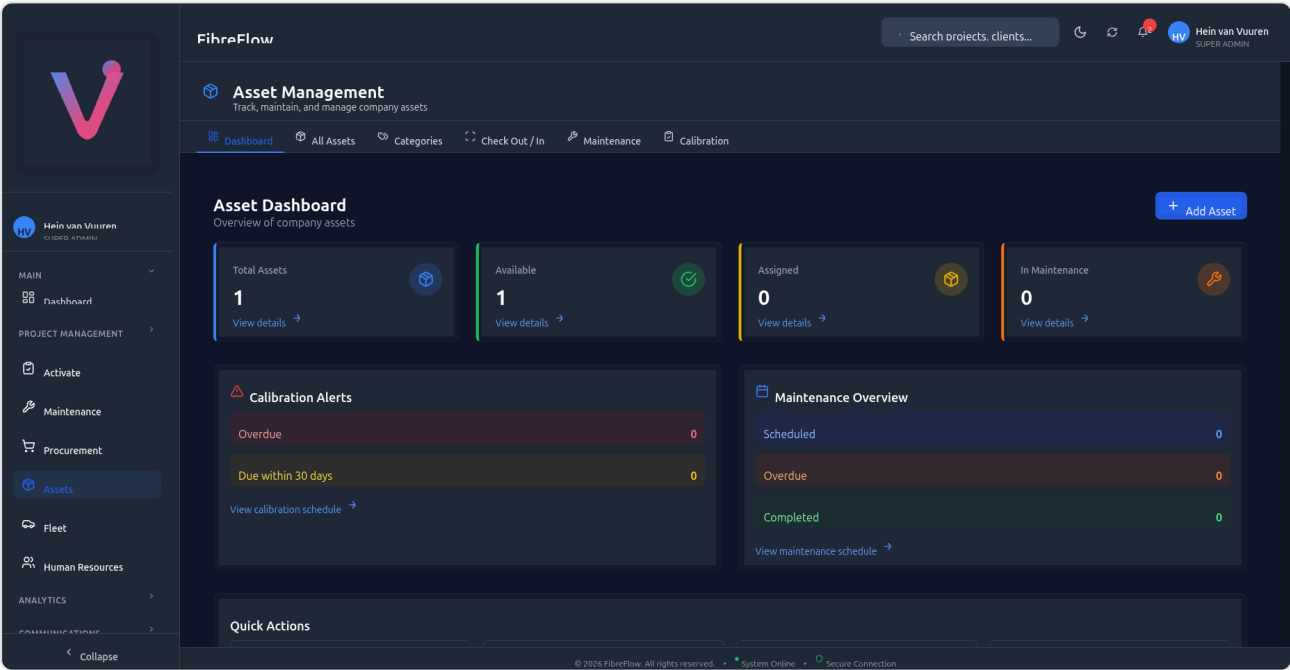


Figure 7.2: Assets list with search, category filter, and status indicators

VIEWING ASSETS

A searchable table of all assets with:

- **Asset Number** — Unique identifier
- **Name** — Asset description

- **Category** — Classification (Tools, Electronics, Vehicles, etc.)
- **Status** — Available, Assigned, In Maintenance, Retired
- **Location** — Current location or assigned person
- **Last Calibration** — Date of last calibration check

CREATING AN ASSET

1. Click **+ New Asset**
2. Fill in required fields:
 - **Name** — Asset description
 - **Category** — Select from categories
 - **Asset Number** — Auto-generated or manual
 - **Serial Number** — Manufacturer serial
 - **Purchase Date** and **Purchase Price**
 - **Location** — Storage location
3. Optionally add:
 - **Barcode** — For QR code scanning
 - **Warranty Expiry** — Warranty end date
 - **Calibration Interval** — Days between calibrations
4. Click **Save**

7.3 Asset Detail

Click any asset to view its full detail page:

- **Overview** — All asset information
- **Documents** — Upload and manage asset documents (manuals, certificates)
- **Maintenance History** — Past and scheduled maintenance records
- **Assignment History** — Who has checked this asset out and when
- **Verification** — VLM-powered label verification (scan asset label to verify match)

7.4 Categories

Navigation: Assets → **Categories** tab

- Create and manage asset categories (e.g., Power Tools, Fiber Optic Equipment, Safety Gear)
- Each category can have custom fields

7.5 Check Out / Check In

Navigation: Assets → **Check Out / In** tab

CHECKING OUT AN ASSET

1. Navigate to the asset or use **Check Out / In** tab
2. Click **Check Out**
3. Select the **Person** receiving the asset
4. Add **Purpose** and **Expected Return Date**
5. Confirm

CHECKING IN AN ASSET

1. Find the checked-out asset
2. Click **Check In**
3. Note the **Condition** on return
4. Add any **Notes** about damage or issues
5. Confirm

7.6 Maintenance Scheduling

Navigation: Assets → **Maintenance** tab

- Schedule preventive maintenance
- Track maintenance history
- Set recurring maintenance intervals
- View upcoming and overdue maintenance

7.7 Calibration Tracking

Navigation: Assets → **Calibration** tab

- View assets with upcoming or overdue calibration

- Record calibration results
- Upload calibration certificates
- Set calibration intervals per asset

Important: Overdue calibrations are flagged prominently on the dashboard. Using uncalibrated equipment may violate compliance requirements.

8. Fleet Management

The Fleet module manages vehicles, drivers, fuel, GPS tracking, and daily check-ins with AI-powered features.

8.1 Dashboard

Navigation: Sidebar → **Fleet** (click section header)

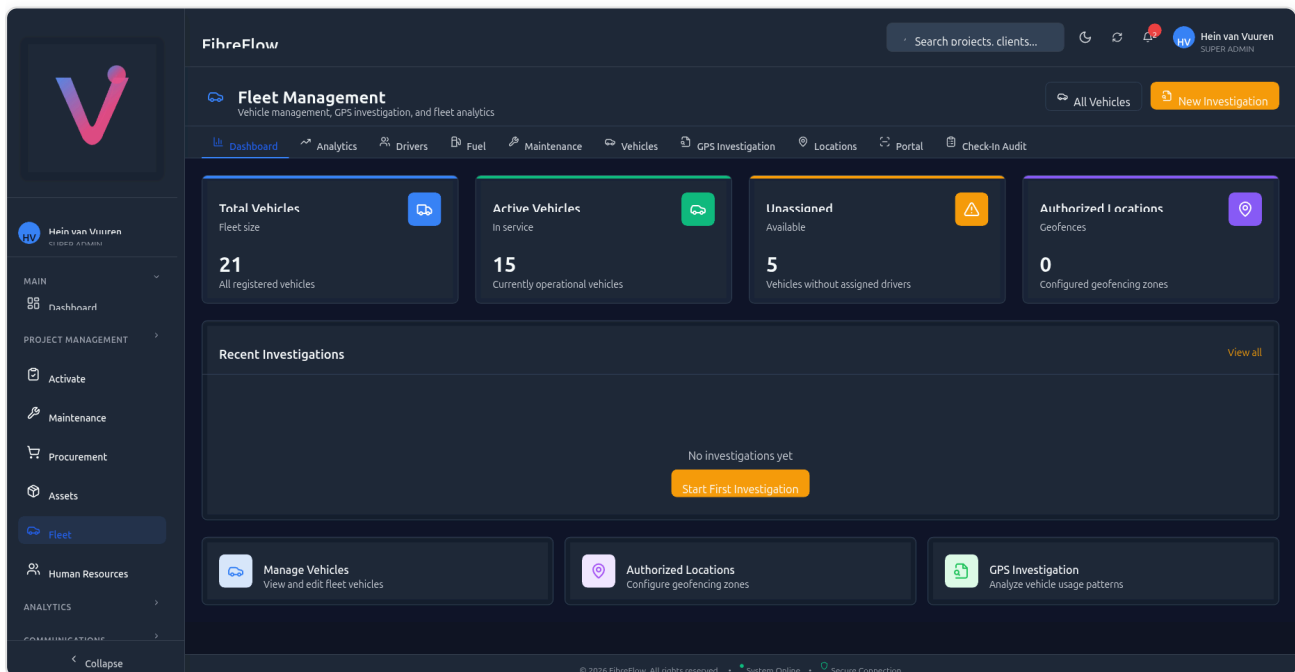


Figure 8.1: Fleet Dashboard showing vehicle count, fuel spend, maintenance alerts, and driver stats

8.2 Vehicles

Navigation: Fleet → Vehicles tab

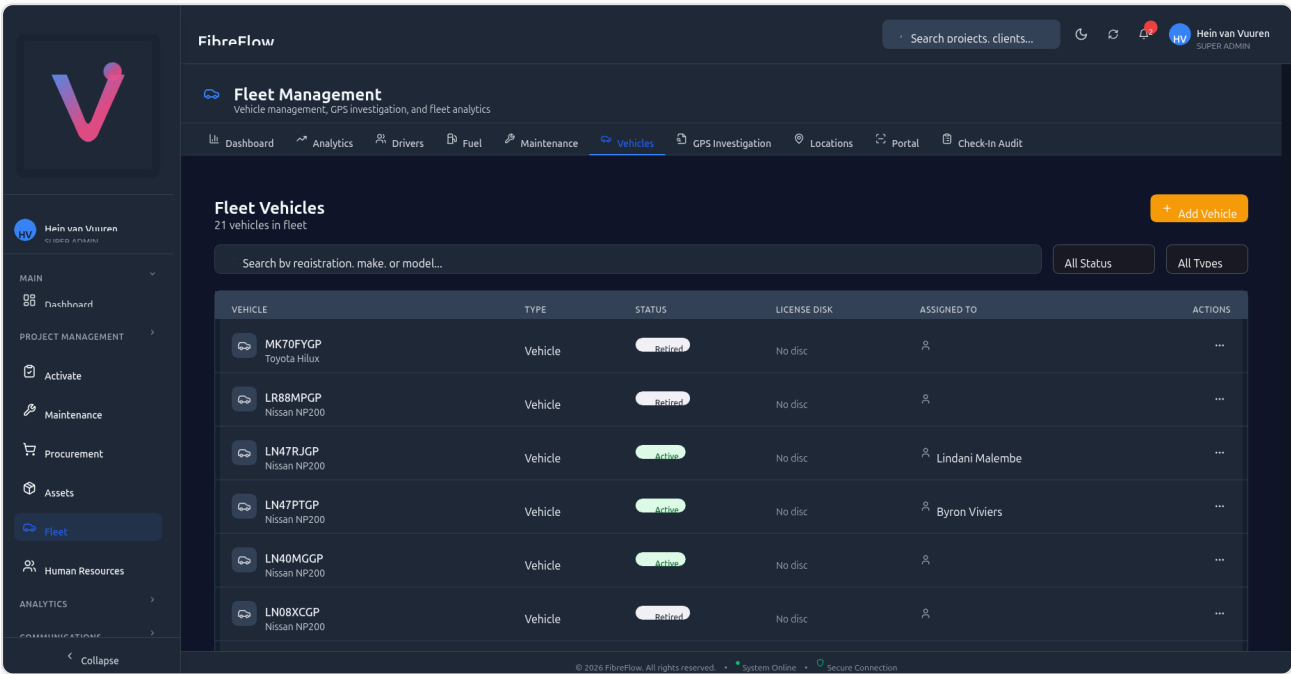


Figure 8.2: Fleet vehicles list showing registration, make, driver assignment, and status

VEHICLE INFORMATION

Each vehicle record includes:

- Registration number and VIN
- Make, model, and year
- Assigned driver
- Current mileage
- Fuel level
- Service due date
- Insurance and license expiry
- Documents (registration, insurance, etc.)

8.3 Drivers

Navigation: Fleet → Drivers tab

- **Driver List** — All registered drivers with license status

- **Driver Detail** — Contact info, assigned vehicle, license details
- **Documents** — Upload and track:
 - Driver's license (front and back)
 - PrDP (Professional Driving Permit)
 - Medical certificate
- **License Expiry Alerts** — System warns when licenses are expiring

8.4 Vehicle Check-In

The daily/weekly vehicle inspection system with VLM-powered features.

Navigation: Fleet → **Check-In Audit** tab (or via Fleet Portal)

CHECK-IN PROCESS

1. **Plate Scan** — The driver photographs the vehicle license plate
2. **VLM Verification** — AI reads the plate and verifies it matches the assigned vehicle
3. **First-Time Calibration** (new vehicles only):
 - Enter current odometer reading
 - Select current fuel level (0-100%)
 - Upload dashboard photo
4. **Inspection Checklist** — Complete the inspection items:
 - Tire condition
 - Lights and indicators
 - Fluid levels
 - Body damage
 - Interior cleanliness
5. **Dashboard Reading** — VLM reads the dashboard photo to extract:
 - Odometer reading
 - Fuel level
 - Warning lights
6. **Photos** — Upload photos of any issues found
7. **Submit** — Complete the check-in

Tip: The VLM (AI) reads license plates and dashboard displays automatically. However, always verify the readings are correct before submitting.

8.5 Fuel Management

Navigation: Fleet → **Fuel** tab

- **Fuel Transactions** — Record fuel purchases with receipt scanning
- **Fuel Summary** — Total fuel spend by vehicle, driver, and period
- **Anomaly Detection** — System flags unusual fuel consumption patterns
- **Fuel Level Tracking** — Historical fuel levels from check-in data

8.6 Fleet Maintenance

Navigation: Fleet → **Maintenance** tab

- Schedule vehicle service and maintenance
- Track service history
- Set recurring maintenance intervals (km-based or time-based)
- View upcoming and overdue service items
- Manage service providers and costs

8.7 GPS Investigation

Navigation: Fleet → **GPS Investigation** tab

- **Upload GPS Data** — Import CSV files from GPS tracking devices
- **Trip Analysis** — View individual trips with routes, stops, and speeds
- **Anomaly Detection** — Flag unauthorized trips, excessive speed, or unusual routes
- **Driver Scorecards** — Performance ratings based on driving behavior

8.8 Locations & Portal

Navigation: Fleet → **Locations / Portal** tabs

- **Locations** — Manage depot and parking locations
- **Portal** — Self-service driver portal for check-ins and reporting

9. Human Resources

The Human Resources module manages staff information, departments, compliance, and organizational structure.

9.1 Staff Directory

Navigation: Sidebar → **Human Resources** (click section header)

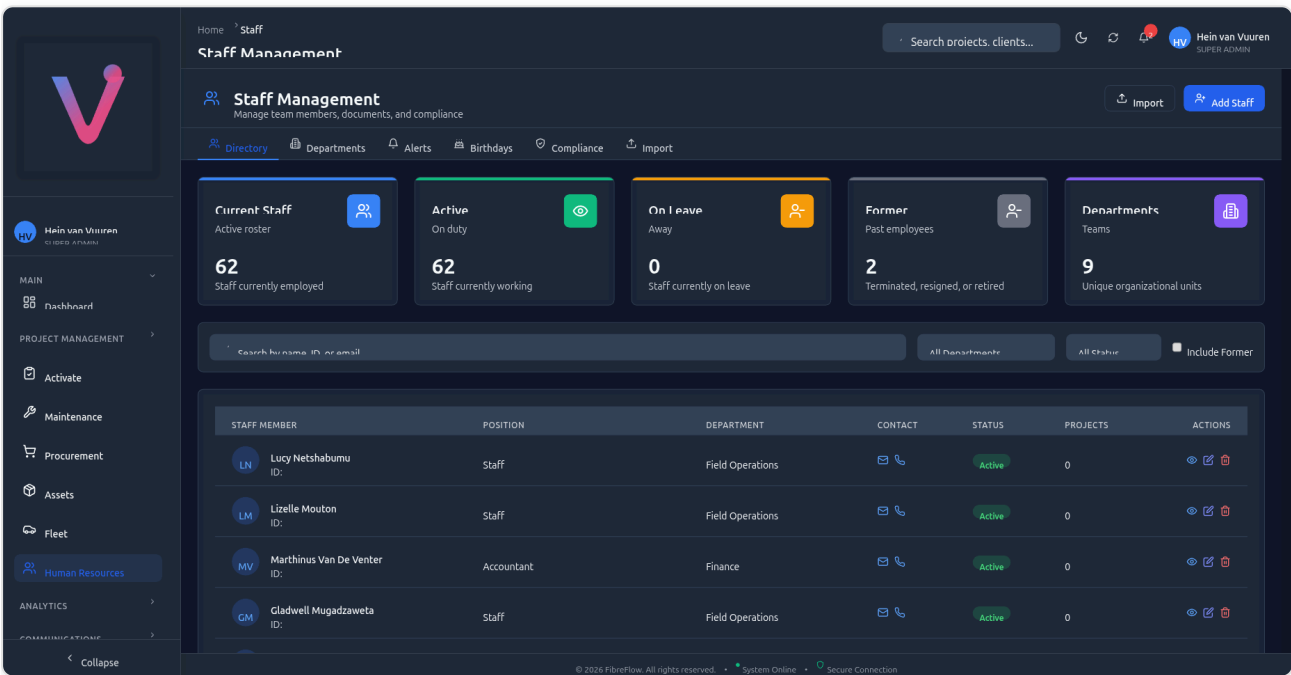


Figure 9.1: Staff Directory showing employee list with search, department filter, and status

FEATURES

- **Search** — Find staff by name, email, or employee number
- **Filters** — Filter by department, position, status (active/inactive)
- **Grid/List View** — Toggle between card and table layouts
- **Quick Actions** — Add new staff member, export list

9.2 Staff Detail

Click any staff member to view their full profile:

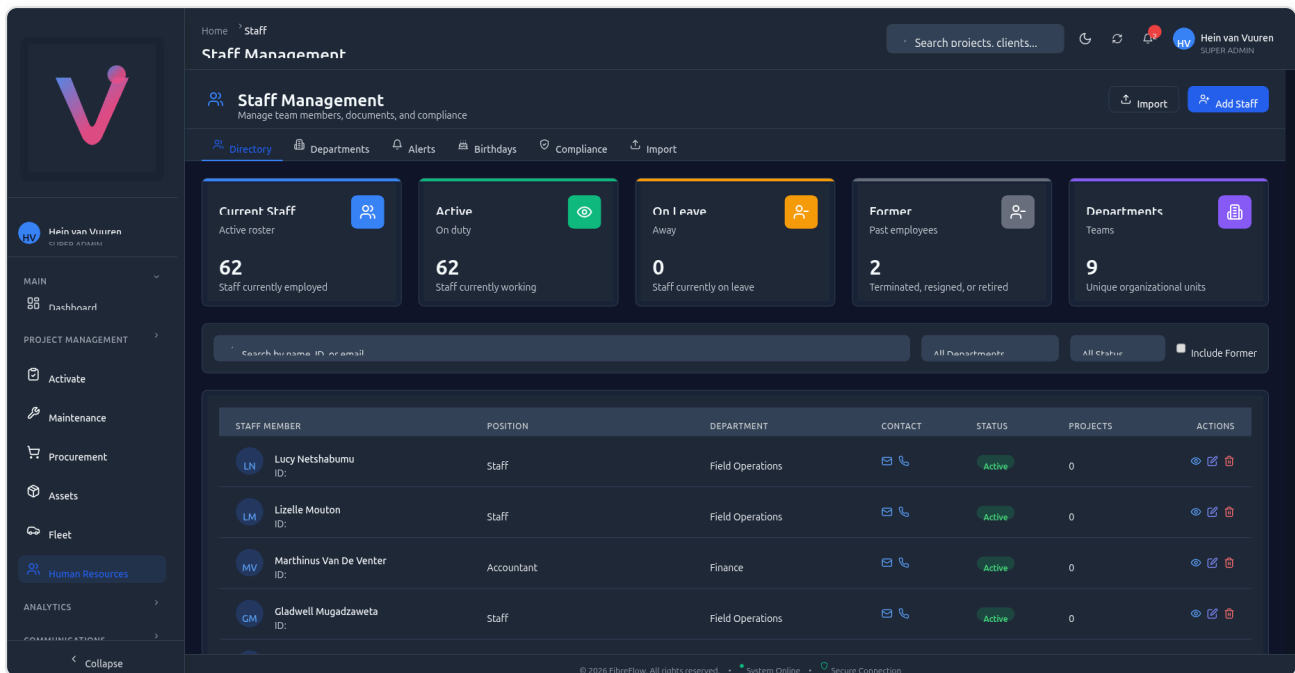


Figure 9.2: Staff detail page showing personal info, employment details, and compliance status

TABS

Tab	Content
Overview	Personal info, contact, address, identity documents
Employment	Position, department, start date, reporting line
Compliance	Document status, training records, certifications
Notes	Internal notes and communications log
Projects	Projects this staff member is assigned to

EDITING STAFF

1. Click **Edit** on the staff detail page
2. The edit form has three sections:
 - **Overview** — Personal info, contact details, next of kin
 - **Employment** — Position, department, salary, contract type
 - **Compliance** — Document uploads and status tracking

3. Make changes and click **Save**

9.3 Departments

Navigation: Human Resources → **Departments** tab

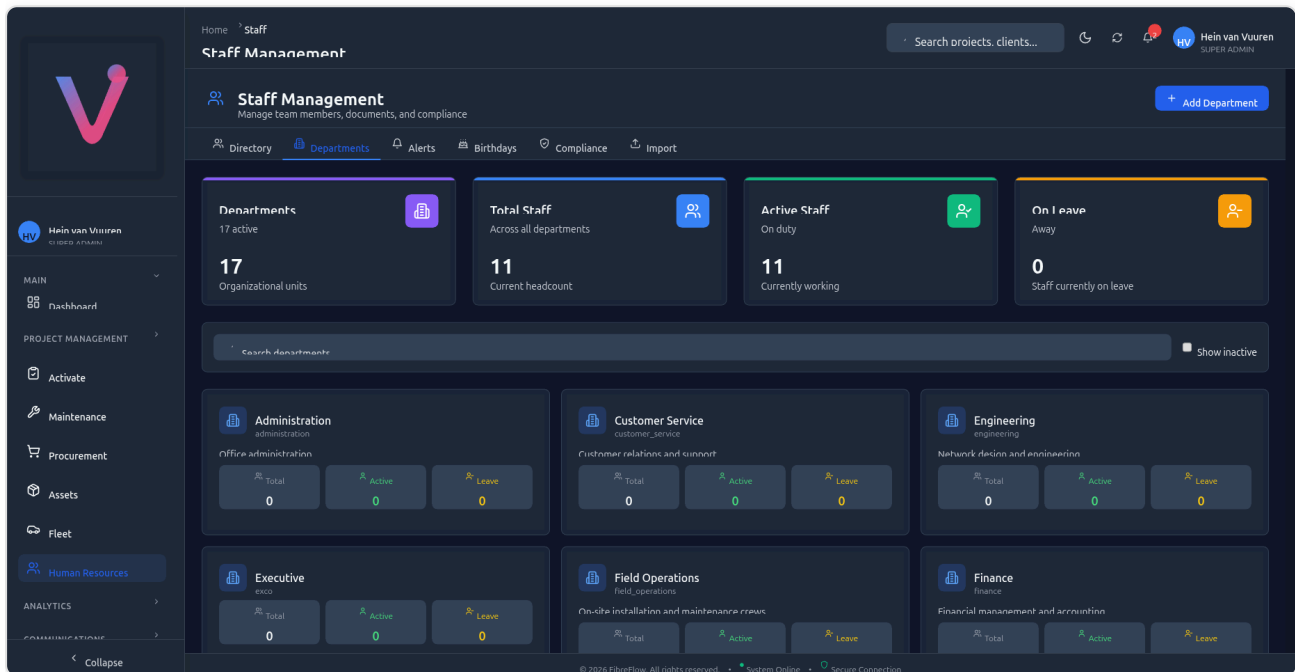


Figure 9.3: Departments grid showing department cards with member counts and status

- View all departments as cards with member count
- Create new departments
- Edit department details
- View department reports (headcount, growth trends)

9.4 Alerts & Birthdays

Navigation: Human Resources → **Alerts / Birthdays** tabs

- **Alerts** — Expiring documents, overdue compliance items, license renewals
- **Birthdays** — Upcoming staff birthdays for team recognition

9.5 Compliance Tracking

Navigation: Human Resources → **Compliance** tab

Track staff document compliance:

- Driver's licenses
- Medical certificates
- Safety training
- Background checks
- Professional certifications

Each document type shows:

- Status (Valid, Expiring, Expired, Missing)
- Expiry date
- Upload date
- Alert thresholds

9.6 Staff Import

Navigation: Human Resources → **Import** tab (**Admin only**)

- **Upload CSV/Excel** — Bulk import staff records
- **Field Mapping** — Map file columns to staff fields
- **Validation** — Preview and fix errors before import
- **Overwrite Option** — Choose whether to update existing records

10. Analytics

The Analytics section provides dashboards and reports for organizational performance insights.

10.1 Analytics Dashboard

Navigation: Sidebar → **Analytics** → **Analytics Dashboard**

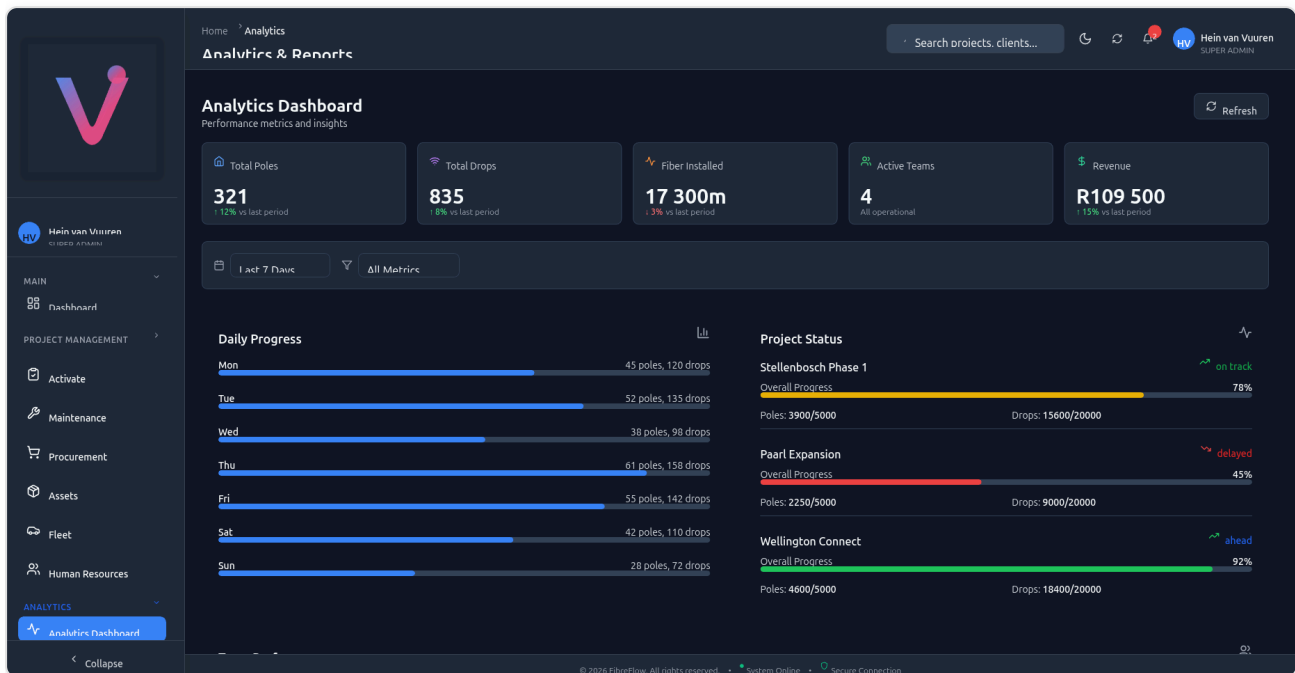


Figure 10.1: Analytics Dashboard showing project metrics, team performance, and trend charts

COMPONENTS

- **Statistics Cards** — Key metrics at a glance (projects, completions, revenue, issues)
- **Daily Progress Chart** — Visual timeline of daily progress across projects
- **Project Status View** — Distribution of projects by status
- **Team Performance Table** — Staff productivity metrics
- **Key Insights** — AI-generated insights and recommendations

10.2 Enhanced KPIs

Navigation: Sidebar → Analytics → Enhanced KPIs

Advanced Key Performance Indicator tracking with:

- Custom KPI definitions
- Target vs actual tracking
- Trend visualization over time
- Drill-down by project, team, or period

10.3 KPI Dashboard

Navigation: Sidebar → **Analytics** → **KPI Dashboard**

A focused view of the most important operational KPIs:

- Installation completion rates
- Maintenance response times
- QA pass rates
- Financial metrics

10.4 Reports

Navigation: Sidebar → **Analytics** → **Reports**

Generate and export operational reports:

- **Project Reports** — Progress, timeline adherence, budget
- **Staff Reports** — Productivity, attendance, compliance
- **Financial Reports** — Revenue, expenses, profitability
- **Operational Reports** — Ticket volumes, SLA compliance

11. Communications

The Communications section manages WhatsApp integration, meeting coordination, and team notifications.

11.1 Communications Portal

Navigation: Sidebar → **Communications** → **Communications Portal**

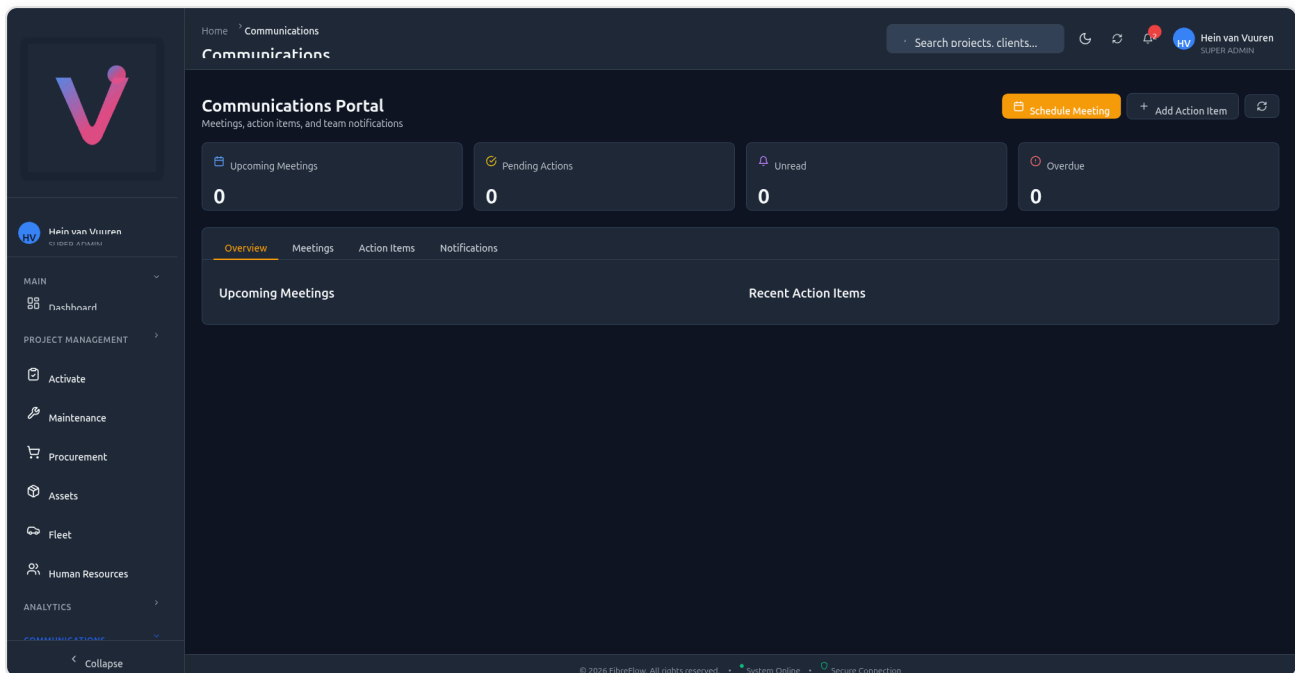


Figure 11.1: Communications Portal showing messaging overview and notification channels

A unified view of organizational communications:

- Overview statistics
- Recent communications
- Notification channels
- Action items from communications

11.2 WhatsApp Administration

(Admin only)

Navigation: Sidebar → **Communications** → **WhatsApp Portal**

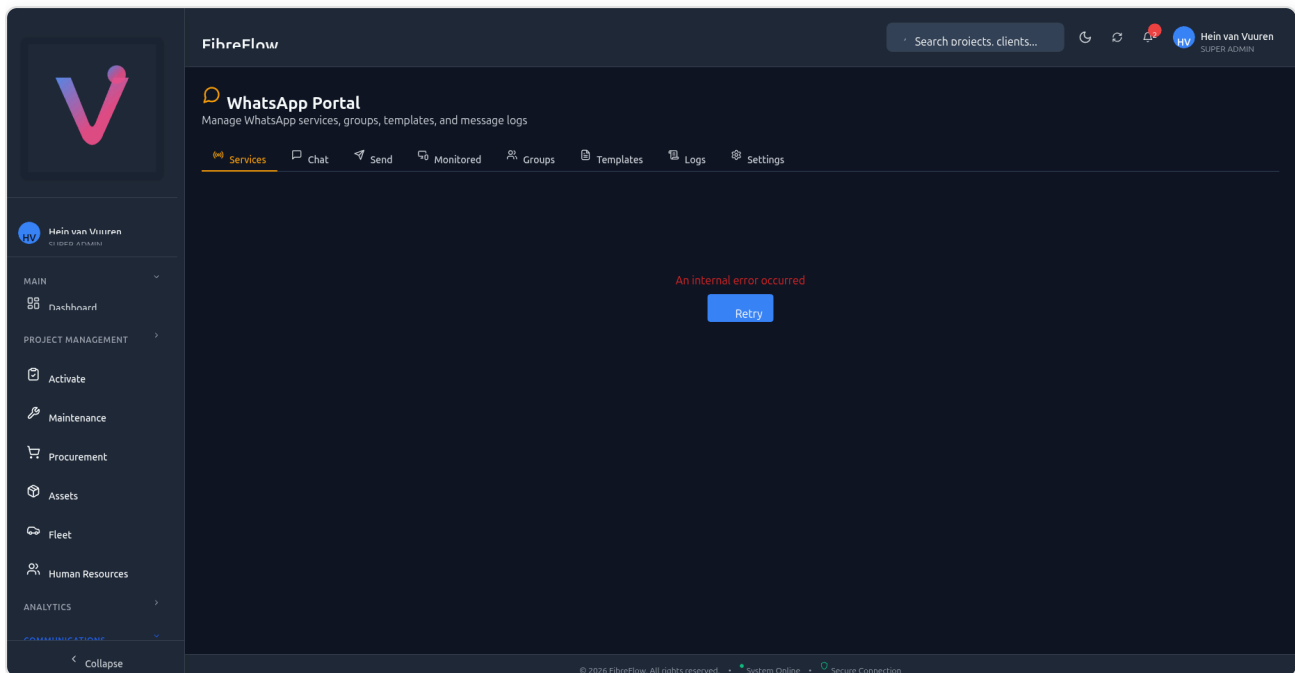


Figure 11.2: WhatsApp Administration showing groups, message logs, and service health

TABS

Tab	Purpose
Groups	Manage WhatsApp project groups
Logs	View message delivery logs with export
Templates	Manage message templates
Services	Monitor WhatsApp service health (Sender, Bridge, Feedback)
Settings	Configure WhatsApp integration settings

KEY FEATURES

- **Group Management** — Create and manage WhatsApp groups per project
- **Message Logs** — Search, filter, and export message history
- **Template Management** — Create reusable message templates
- **Service Health** — Monitor the health of WhatsApp infrastructure:
 - **Sender** (port 8081) — Sends outgoing messages
 - **Bridge** (port 8083) — Receives DR submissions

- **Feedback** (port 8092) — Sends QA feedback

11.3 PDF Tools

Navigation: Sidebar → **Communications** → **PDF Tools**

A collection of PDF manipulation tools:

- Merge PDFs
- Split PDFs
- Convert documents to PDF
- Compress PDFs
- Rotate pages
- Add watermarks

12. System Administration

The System section provides administrative tools for monitoring, data management, and configuration.

(Admin only — most features)

12.1 System Health Hub

Navigation: Sidebar → **System** → **System Health Hub**

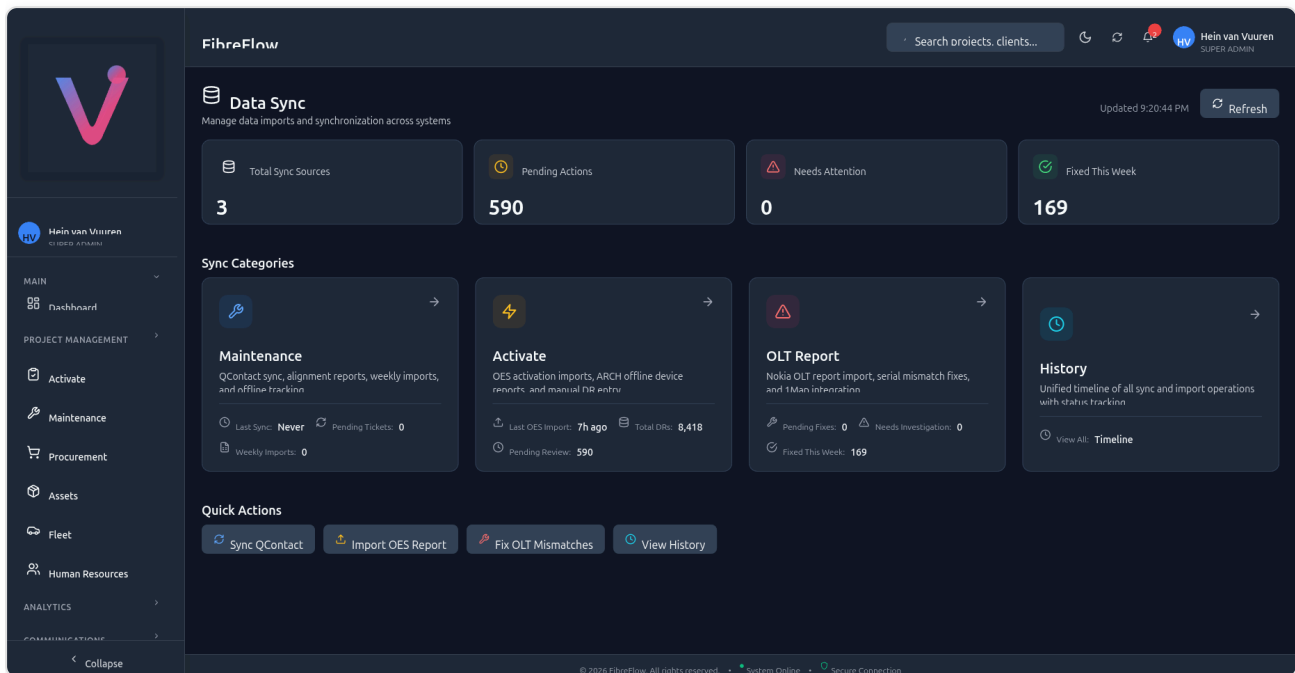


Figure 12.1: System Health Hub showing service status, API health, and recent alerts

Monitor the health of all FibreFlow services:

- **API Health** — Status of all API endpoints
- **Database** — Connection pool status and query performance
- **External Services** — VLM, QContact, Smartsheet, OneMap connectivity
- **Job Queue** — Background task processing status
- **Error Tracking** — Recent errors and their frequency

12.2 Data Sync

Navigation: Sidebar → **System** → **Data Sync**

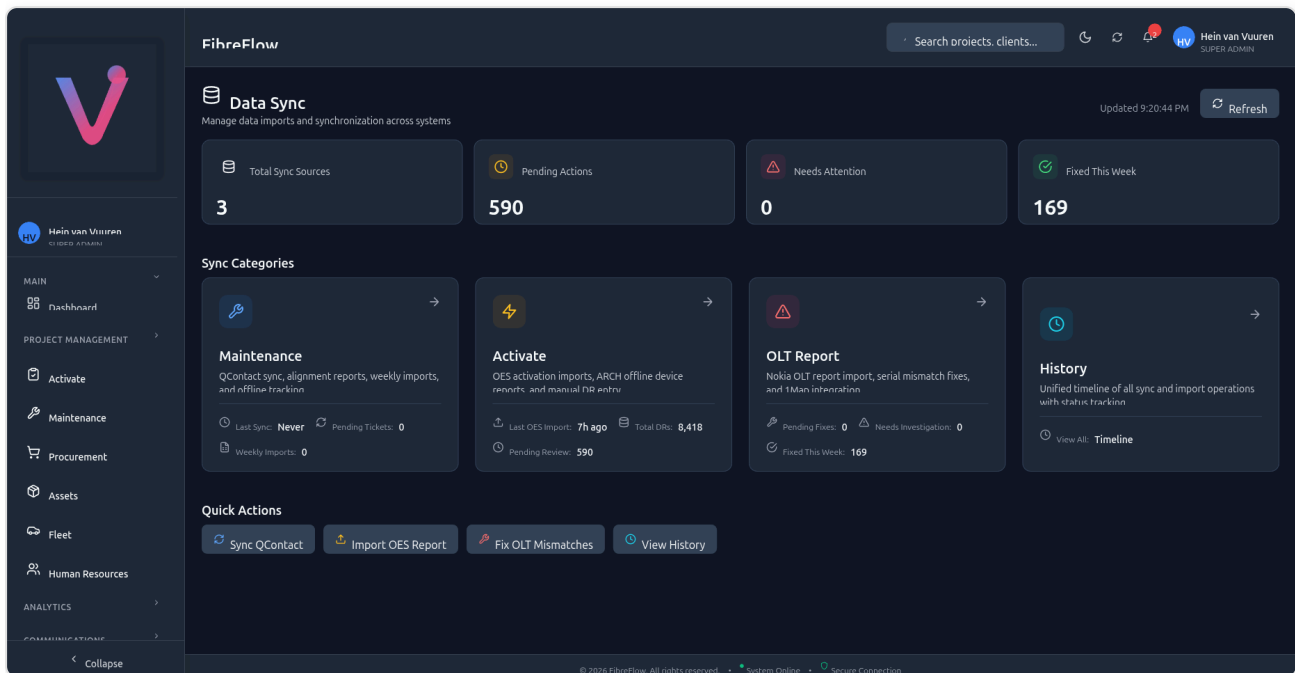


Figure 12.2: Data Sync page showing sync operations, OLT report, and operation history

Manage data synchronization across systems:

SYNC GROUPS

Group	Description
OLT Report	Import and fix ONT serial mismatches from OLT reports
OES Sync	Import OES activation data from Excel
SharePoint Sync	Sync DR data from SharePoint
QField Sync	Push coordinates to QField for field mapping
1Map Sync	Sync serial numbers with 1Map (OneMap)
QContact Sync	Maintenance ticket synchronization

OLT REPORT (SPECIAL FEATURE)

The OLT Report tool manages ONT serial number corrections:

1. Import OLT mismatch report (Excel)
2. View mismatched records (wrong serial in 1Map)

3. **Auto-fix** — System corrects serials in 1Map via API
4. **Investigate** — Records needing manual lookup
5. **Reporting** — Track fixes by date, status, and import batch

12.3 Infrastructure

Navigation: Sidebar → **System** → **Infrastructure**

Monitor FibreFlow infrastructure:

- Server status (Production, Staging, Dev)
- Service health checks
- Resource utilization
- Links to external monitoring (xyOps, Grafana)

12.4 Settings

Navigation: Sidebar → **System** → **Settings**

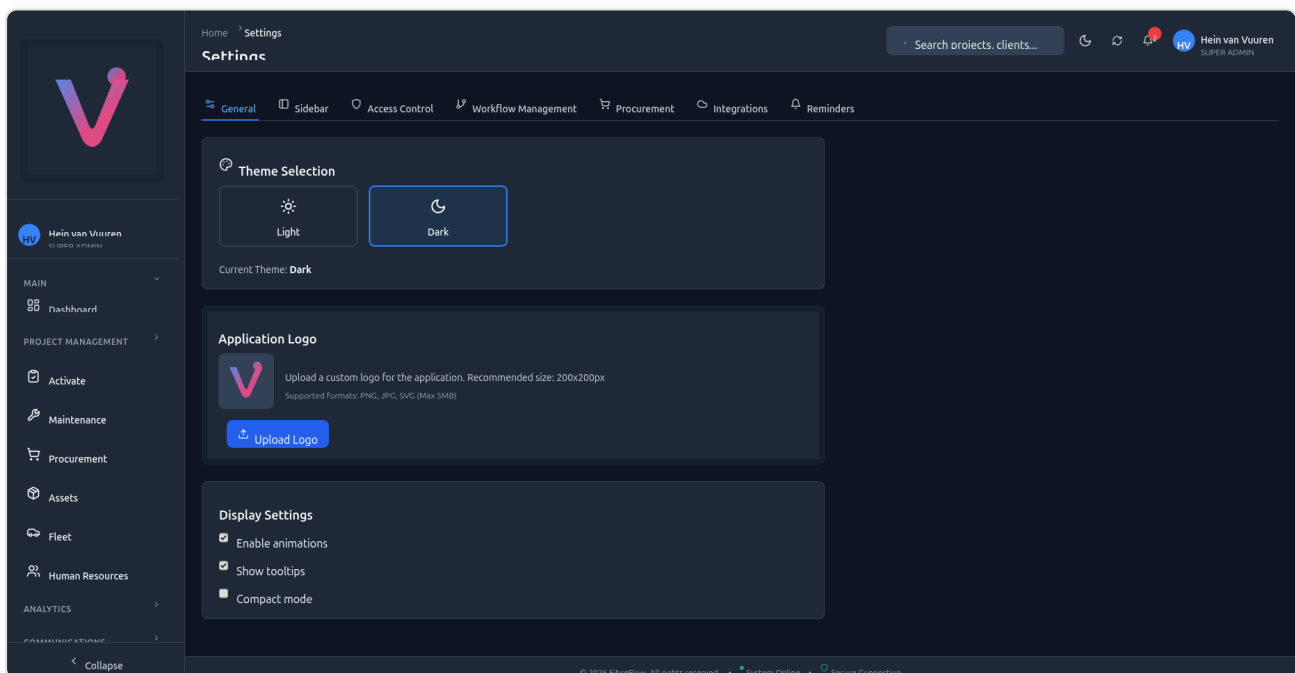


Figure 12.3: Settings page showing organizational, procurement, and user preference tabs

SETTINGS TABS

Tab	Content
General	Organization name, branding, defaults
Procurement	Approval thresholds, default terms, tax rates
User Preferences	Sidebar customization, theme, notifications
Positions	Manage staff positions
Departments	Manage department list
Hierarchy	Reporting structure configuration

SIDEBAR CUSTOMIZATION

Users can customize their sidebar by:

1. Go to **Settings** → **User Preferences**
2. Select which sections to show/hide
3. Reorder sections by drag-and-drop
4. Save preferences (stored per user)

12.5 Imports & Downloads

Navigation: Sidebar → **System** → **Imports / Downloads**

- **Imports** — SOW document import with data extraction
- **Downloads** — Export templates, reports, and data extracts

13. Appendices

Appendix A: Role Permissions Matrix

Module	Super Admin	Admin	Manager	Technician	Viewer	Contractor
Dashboard	Full	Full	Full	View	View	—
Projects	Full	Full	Full	View	View	Assigned
Clients	Full	Full	Full	—	View	—
Contractors	Full	Full	Full	—	View	Own
Activate	Full	Full	Full	—	View	—
Maintenance	Full	Full	Full	Assigned	View	—
Procurement	Full	Full	Full	—	View	—
Assets	Full	Full	Full	Checkout	View	—
Fleet	Full	Full	Full	Own	View	—
HR / Staff	Full	Full	View	Own	—	—
Analytics	Full	Full	Full	—	View	—
Communications	Full	Full	View	—	View	—
WhatsApp Admin	Full	Full	—	—	—	—
System Admin	Full	—	—	—	—	—
Settings	Full	Full	Limited	—	—	—

Legend: Full = All CRUD operations | View = Read only | Assigned = Only assigned items | Own = Only own records | — = No access

Appendix B: Keyboard Shortcuts

Shortcut	Action
Tab	Navigate between form fields
Shift+Tab	Navigate backwards between fields
Escape	Close modal or cancel action
Ctrl+Enter	Submit form
Ctrl+K	Open global search (where available)
/	Focus search bar (on list pages)

Appendix C: Glossary of Terms

Term	Definition
BOQ	Bill of Quantities — itemized list of materials for a project
DR	Drop Reference — unique identifier for a fiber drop point (e.g., DR1738371)
FT Number	FiberTime reference number from QContact (e.g., FT695563)
GRN	Goods Receipt Note — document confirming receipt of ordered items
KPI	Key Performance Indicator — measurable metric for tracking performance
OES	Operation Engineering Support — Nokia activation data
OLT	Optical Line Terminal — equipment at the service provider end
ONT	Optical Network Terminal — customer premises equipment
PO	Purchase Order — formal order to a supplier
PON	Passive Optical Network — fiber distribution network
QA	Quality Assurance — verification process for installations
QContact	FiberTime's CRM system (fibertime.qcontact.com)
QField	Mobile GIS application for field mapping
RBAC	Role-Based Access Control — permission system
RFQ	Request for Quotation — invitation for suppliers to bid
SLA	Service Level Agreement — response time target
SOW	Scope of Work — project scope document defining work
VLM	Vision Language Model — AI for analyzing photos
VPS	Virtual Private Server — remote server infrastructure

Appendix D: System Architecture

ENVIRONMENTS

Environment	URL	Purpose
Production	app.fibreflow.app	Live system
Staging	vf.fibreflow.app	Pre-release testing
Development	dev.fibreflow.app	Feature development
Backup	backup.fibreflow.app	Redundancy

EXTERNAL INTEGRATIONS

System	Purpose	Sync Method
QContact	Maintenance CRM	API (bidirectional)
Smartsheet	Pipeline tracking	API (import)
Fireflies.ai	Meeting transcription	API (import)
1Map / OneMap	GIS data (drops, photos)	API (bidirectional)
QField	Mobile GIS mapping	GeoJSON push
Odoo	ERP (inventory, POs)	API (import)
Firebase	File storage	SDK
SharePoint	Document sync	API (import)

Appendix E: Troubleshooting Guide

GENERAL ISSUES

Problem	Solution
Page not loading	Refresh the page. Check your internet connection. Try clearing browser cache.
"Unauthorized" error	Your session may have expired. Log out and log back in.
Missing menu items	Your role may not have access. Contact your administrator.
Data not updating	Try refreshing the page. Some data syncs on intervals (e.g., QContact).
Slow performance	Check your internet speed. Try a different browser. Close unnecessary tabs.

MODULE-SPECIFIC ISSUES

Module	Problem	Solution
Activate	Photos not loading	Check if the 1Map API is responding. Photos require VPS connectivity.
Activate	VLM categorization wrong	Correct the categorization manually — corrections improve future accuracy.
Maintenance	QContact sync failed	Check Data Sync tab for error details. Try manual sync.
Maintenance	Cannot move ticket status	Check required fields (fault cause, verification). Verify RBAC permissions.
Procurement	BOQ import fails	Ensure Excel has headers in row 1. Check column mapper for correct mappings.
Procurement	PO stuck in approval	Check Approvals tab. The assigned approver may need to take action.
Fleet	Plate scan not recognized	Ensure good lighting and clear plate photo. Enter plate manually if needed.
Assets	Cannot delete asset	Assets must be checked in (status: Available) before deletion.
HR	Import errors	Check CSV formatting. Required fields: first_name, last_name, email.

This manual is maintained by the Velocity Fibre development team. For questions, corrections, or feature requests, contact the system administrator.

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